A Principled and Interpretable Orbit-Based Learning Framework

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Editor:

Abstract

We propose a structural theory of learnability from an orbit-based operator-theoretic perspective in which learnability is organized by dataset-induced orbit geometry and constructive realizability rather than by unrestricted parametrization alone. We develop this perspective in finite dimensions, while showing that its core constructive mechanisms already persist in infinite-dimensional Banach settings.

1 Introduction

Deep learning excels across vision, language, and time series, but much of its practical success still relies on stacked nonlinearities, heavy overparameterization, and architectural heuristics such as attention, which often obscure mechanism and demand large labeled datasets and careful tuning. In this work we ask a different question: can learning be organized around simple orbit dynamics, explicit geometry, and constructive realization rather than around increasingly opaque parametrization? We introduce an orbit-based framework in which scaffold geometry, realizability, and prediction are treated as three linked structural bottlenecks rather than as disconnected stages of a pipeline. The predictive instantiation of this framework, which we call a Weighted Backward Shift Neural Network (WBSNN), has a general-purpose design: the same orbit-based mechanism can be deployed across varied and demanding settings without requiring the pipeline to be tailored to each dataset. Moreover, although the backbone is linear, the model's predictions become nonlinear through their dependence on orbit evolution and data-dependent coefficient selection, yielding both interpretability and expressive capacity.

The philosophy is the following. Instead of taking the model as the primary object and asking it to discover structure implicitly, we first extract from the data a structured orbit scaffold encoding part of the dataset's intrinsic geometry. On that scaffold we study whether a target operator, or in the predictive setting a target dataset map, can be constructively realized with explicit and analyzable error. Prediction is then built on top of this constructive stage: rather than learning outputs freely, the predictor learns a coefficient law that tracks and extends the constructive rule found on the training set. This leads to a framework organized around three explicit bottlenecks: a *scaffold bottleneck*, measuring how well the selected orbit scaffold captures the relevant geometry of the data; a *realizability bottleneck*, measuring how well the projected target can be constructively synthesized on that scaffold; and a *predictive coefficient bottleneck*, measuring how well the learned predictor reproduces and generalizes the constructive coefficient rule.

The theoretical contribution of the paper has two tightly connected parts. On the realizability side, we prove exact interpolation theorems showing that orbit-generated realizations exist under explicit structural conditions; we derive a two-term decomposition separating projection and parametrization errors; we develop an algorithm that searches for an optimal orbit-constrained scaffold; and we introduce CCDS, a constructive deviation-controlled synthesis mechanism that yields explicit certificates and recovers classical greedy pursuit as a boundary case. On the predictive side, we show that the same structural logic carries over to learning: the WBSNN predictor inherits the scaffold and realizability bottlenecks and adds only one new one, namely the learned approximation of the effective CCDS coefficient map. This yields in-sample structural prediction bounds and an out-of-sample transfer principle under geometric proximity assumptions between training and test points. In summary, the paper should be read as a single theory of learnability: Section 2 develops its constructive realizability backbone, and Section 3 shows that prediction is obtained by extending the same scaffold-realization logic rather than by introducing a separate model class.

Why should the machine-learning community care about such a framework? Because it shifts the focus of learnability away from opaque feature mixing and unrestricted parametrization, and toward a more explicit structural question: what geometry the dataset induces, what targets can be constructively realized on that geometry, and how prediction can be built by extending that constructive rule. In this sense, the framework is interpretable end-to-end: the scaffold is data-induced and geometric, the realization step is explicit and constructive, and prediction is constrained by orbit coefficients rather than delegated to a black-box parametrization. More fundamentally, the framework suggests that datasets should not be viewed merely as collections of samples to be fit, but as objects carrying intrinsic geometric structure that directly shapes what can be learned from them. A single running example of this geometry-driven view is provided in subsection 4.1, where near-collinear orbit atoms expose the necessity of deviation-controlled synthesis (CCDS) over classical greedy pursuit.

Our construction is rooted in the theory of linear dynamics, where the weighted backward shift operator plays a foundational role. In infinite-dimensional separable sequence Banach spaces, this operator acts on sequences $x = (x_0, x_1, x_2, \dots)$ by shifting coordinates backward and scaling by a bounded weight sequence $w = (w_i)_{i \geq 0}$:

$$B_w x = (w_1 x_1, w_2 x_2, w_3 x_3, \dots). \tag{1}$$

The orbit of a point x under B_w is the sequence $\mathrm{Orb}(x, B_w) = \{B_w^m x : m \geq 0\}$, that is, the collection of all iterates generated by repeatedly applying the shift. This simple mechanism gives rise to rich dynamical behavior, including dense orbits and hypercyclicity, and is therefore a natural source of structured representations. To adapt this dynamics to finite-dimensional learning problems, we introduce a wrapped finite-dimensional surrogate that cyclically reintroduces lost coordinates while preserving the shift-based orbit structure. This weighted backward shift mechanism serves as the backbone of the entire framework: it generates the orbit dictionary used to build scaffolds, supports the exact interpolation theory underlying realizability, and provides the structured representation class on which WBSNN performs prediction.

In this sense, the paper shifts the center of gravity of learning theory from parametrization alone to the interaction between dataset geometry, orbit structure, and constructive realization.

The framework's backbone. At the core of our framework lies a structured, shift-based architecture $W^{(L)}$ (introduced and defined below), which cyclically shifts and scales input features across layers. Unlike traditional feedforward networks, its modulo-d design preserves all input dimensions throughout depth, enabling persistent feature recombination. This recurrence serves as a finite-dimensional surrogate of orbit dynamics from infinite-dimensional operator theory, where iterated linear actions generate structured, dense trajectories. With only d parameters, $W^{(L)}$ builds global representations from local shifts, providing an interpretable and expressive backbone for both interpolation and generalization.

Definition 1 *We introduce the* wrapped *finite-dimensional iterate, denoted $W^{(m)}$, used as a technical surrogate for backward-shift dynamics in infinite dimension: rather than shifting mass out of the ambient space, indices are taken modulo d, so that orbit iterates remain in $\mathbb{R}^d$ while still exhibiting* *shift-like propagation. Concretely, for $x \in \mathbb{R}^d$ and weights $w = (w_0, \ldots, w_{d-1}) \in \mathbb{R}^d$, we set*

$$\left(W^{(m)}x\right)_{i-1} := \left(\prod_{k=0}^{m-1} w_{(i+k) \bmod d}\right) x_{(i+m-1) \bmod d}, \tag{2}$$

for $i = 1, \ldots, d$ and $m \geq 1$, with the empty product interpreted as 1 for $m = 0$.

This wrapping is not a standard finite-dimensional dynamical system; rather, it is a technical device that adapts shift-based orbit constructions, naturally defined in infinite-dimensional spaces, to a fixed finite dimension.

Note. For intuition, the reader may view $W^{(L)}x$ as a shift-and-scale transformation of x, beginning at the L-th entry and preserving the cyclic order of the components modulo d.

Example 1 *Let $d = 4$, $L = 2$, and weights (w_0, w_1, w_2, w_3). Then for an input vector $x = (x_0, x_1, x_2, x_3) \in \mathbb{R}^4$, then*

$$W^{(2)}x = \begin{bmatrix} 0 & 0 & w_1w_2 & 0 & 0 & 0 \\ 0 & 0 & 0 & w_2w_3 & 0 & 0 \\ 0 & 0 & 0 & 0 & w_3w_0 & 0 \\ 0 & 0 & 0 & 0 & 0 & w_0w_1 \end{bmatrix} \begin{pmatrix} x_0 \\ x_1 \\ x_2 \\ x_3 \\ x_0 \\ x_1 \end{pmatrix} = \begin{pmatrix} w_1w_2x_2 \\ w_2w_3x_3 \\ w_3w_0x_0 \\ w_0w_1x_1 \end{pmatrix}.$$

Remark 2 *The modulo-d wrap-around prevents the iterates $W^{(L)}$ from vanishing as in the classical finite-dimensional weighted backward shift. Hence, unlike the classical case, iteration does not destroy coordinates, but instead propagates and recombines information across all positions. Thus the wrapped shifts provide a structured linear backbone in which information can circulate across all coordinates under iteration, with low parameter complexity.*

Example 2 (Weight regimes $(d = 3, L = 2)$) *Let $w = (c, c, c)$, then $W^{(2)}x = \left(c^2x_2,\ c^2x_0,\ c^2x_1\right)$. Thus, $c = 0.5$ (vanishing) $\Rightarrow$ decay by 0.5^2; $c = 1$ (neutral) $\Rightarrow$ pure permutation (no scaling); $c = 1.5$ (exploding) $\Rightarrow$ amplification by 1.5^2. In general, the product $\prod_{k=0}^{d-1} w_k$ governs energy growth/decay, providing a simple knob for stability and expressivity.*

The following lemma reveals a key structural property of W that limits the distinctiveness of large powers.

Lemma 3 *Given W with d predictors, for every $X \in \mathbb{R}^d$ and every $L \geq 1$, we have that $W^{(L)}X = \lambda W^{(L \bmod d)}X$, where $\lambda = \left(\prod_{k=0}^{d-1} w_k\right)^m$ with $L = md + (L \mod d)$.*

Proof For $1 \leq i \leq d$, (2) gives $(W^{(L)}X)_{i-1} = \left(\prod_{k=0}^{L-1} w_{(i+k) \bmod d}\right) X_{(i+L-1) \bmod d}$. Similarly,

$$(W^{(L \bmod d)}X)_{i-1} = \left(\prod_{k=0}^{(L \bmod d)-1} w_{(i+k) \bmod d}\right) X_{(i+(L \bmod d)-1) \bmod d}.$$

Since $(i + L - 1) \bmod d = (i + (L \bmod d) - 1) \bmod d$, the indices match, so $X_{(i+L-1) \bmod d} = X_{(i+(L \bmod d)-1) \bmod d}$. For $L \geq 1$, there exists $m \geq 0$ such that $L = md + (L \bmod d)$, where $0 \leq L \bmod d < d$. The product in $W^{(L)}$ splits as:

$$\prod_{k=0}^{L-1} w_{(i+k) \bmod d} = \left(\prod_{n=0}^{m-1} \prod_{k=0}^{d-1} w_{(i+nd+k) \bmod d}\right) \prod_{k=0}^{(L \bmod d)-1} w_{(i+md+k) \bmod d}. \tag{3}$$

Indeed, the first md terms of $\prod_{k=0}^{L-1} w_{(i+k) \bmod d}$ are m full cycles (of length d) with indices $i+nd+k$ for n from 0 to $m-1$ and k from 0 to $d-1$. So, for each n, we get

$$w_{(i+nd) \bmod d} \cdot w_{(i+nd+1) \bmod d} \cdot \dots \cdot w_{(i+nd+(d-1)) \bmod d}$$

which covers one full cycle of indices. The remaining $L \bmod d$ terms (from $k = md$ to $k = L-1$) are

$$w_{(i+md) \bmod d} \cdot w_{(i+md+1) \bmod d} \cdot \dots \cdot w_{(i+md+(L \bmod d)-1) \bmod d}$$

which is exactly $\prod_{k=0}^{(L \bmod d)-1} w_{(i+md+k) \bmod d}$.

For each n, the inner product $\prod_{k=0}^{d-1} w_{(i+nd+k) \bmod d} = \prod_{k=0}^{d-1} w_{(i+k) \bmod d} = \prod_{k=0}^{d-1} w_k$, since the indices $(i+nd+k) \bmod d$ cycle through $i, i+1, \dots, i+d-1$. Thus, the first factor in parentheses of the right-hand side of (3) is $\left(\prod_{k=0}^{d-1} w_k\right)^m$. Hence, $(W^{(L)}X)_{i-1} = \left(\prod_{k=0}^{d-1} w_k\right)^m (W^{(L \bmod d)}X)_{i-1}$, for all $i = 1, \dots, d$. Therefore, the scalar $\lambda = \left(\prod_{k=0}^{d-1} w_k\right)^m$ satisfies the lemma. ■

2 Orbit-Based Constructive Realization of Linear Operators

This section develops the constructive backbone of the paper. The central question is whether a target operator can be realized, with explicit and analyzable error, on a data-induced orbit scaffold extracted from the dataset itself. The objects introduced here—scaffold, projected target, projection error, parametrization error, and CCDS realization—will remain the governing objects of the predictive theory in Section 3.

From the viewpoint adopted here, the scaffold is not an auxiliary representation layer but the primary geometric object through which learnability is organized. More precisely, the data first determines an orbit scaffold encoding part of its intrinsic geometry. The target is then projected onto that scaffold, and the resulting projected target is constructively synthesized there. This immediately separates two structural bottlenecks: a geometric bottleneck, measuring how well the scaffold captures the target's action on the data, and a constructive bottleneck, measuring how well the projected target can be realized once the scaffold has been fixed. The remainder of this section develops these two bottlenecks in order: scaffold selection first, then constructive realization on the selected scaffold. Given any linear operator $A : \mathbb{R}^d \to \mathbb{R}^C$, a finite input–target dataset

$$\Xi = \{(x_i, y_i)\}_{i=1}^n \subset \mathbb{R}^d \times \mathbb{R}^C,$$

a fixed dictionary $\mathcal{D} \subset \mathbb{R}^C$, and a tolerance $\varepsilon > 0$, we ask whether there exists a map $\hat{y} : \mathbb{R}^d \to \mathbb{R}^C$ with

$$\{\hat{y}(x) : (x, y) \in \Xi\} \subset \operatorname{span}(\mathcal{D})$$

such that

$$\|A(\cdot) - \hat{y}(\cdot)\|_\Xi < \varepsilon,$$

where the empirical mean seminorm on Ξ is defined by $\|F\|_\Xi := \frac{1}{n}\sum_{i=1}^n \|F(x_i)\|_2$.

This manuscript initiates a principled study of this realizability problem in the case where the dictionary $\mathcal{D}$ is not arbitrary, but is instead a structured, data-induced orbit scaffold encoding the intrinsic geometry of Ξ, as formalized below.

The following is the foundational layer of the proposed framework. Thanks to Hahn-Banach Theorem, it shows that, under a technical independence condition (not hard to show in practice), any finite dataset admits an exact realization within a class of transformed weighted shifts. We now present and prove an explicit construction of such a realization, which will serve as the foundation for our learning algorithm.

Theorem 4 (Exact Interpolation) *Let W on $\mathbb{R}^d$ with weights $w_n \neq 0$, and let $\{(x_i, y_i)\}_{i=1}^{n_0} \subseteq \Xi_{\text{subset}} \subset \mathbb{R}^d \times \mathbb{R}^C$ with $n_0 \leq d$. Suppose for each $i = 1, \ldots, n_0$, there exists $L_i \in \{0, \ldots, d-1\}$ such that*

$$W^{(L_i)} x_i \notin \text{span}\{W^{(L_1)} x_1, \ldots, W^{(L_{i-1})} x_{i-1}\}. \tag{4}$$

Then, there exists a linear operator $J : \mathbb{R}^d \to \mathbb{R}^C$ such that $y_i = JW^{(L_i)} x_i$ for all $i = 1, \ldots, n_0$.

Proof Let W on $\mathbb{R}^d$ with weights $w_n \neq 0$. Let also $(x_i, y_i)_{i=1}^{n_0} \subseteq \Xi_{\text{subset}} \subset \mathbb{R}^d \times \mathbb{R}^C$ satisfying (4).

Initialize: Set $J_0 = 0 : \mathbb{R}^d \to \mathbb{R}^C$.

Iterative construction: For $i = 1$. Pick any $L_1 \in \{0, \ldots, d-1\}$ and find a linear functional $f_1^* \in (\mathbb{R}^d)^*$ with $f_1^*(W^{(L_1)} x_1) = 1$. Define $J_1 v := J_0 v + (y_1 - J_0 W^{(L_1)} x_1) \otimes f_1^*(v) = 0 + (y_1 - 0) \otimes f_1^*(v)$. Thus, $J_1 W^{(L_1)} x_1 = y_1 \cdot 1 = y_1$.

For $i \geq 2$: Assume

$$J_{i-1} W^{(L_j)} x_j = y_j \quad \text{for } j \leq i-1, \tag{5}$$

and pick $L_i \in \{0, \ldots, d-1\}$ satisfying condition (4). By Hahn-Banach Theorem (see Corollary 3.15 on p.112 (8)), there exists a linear functional $f_i^* \in (\mathbb{R}^d)^*$ such that $f_i^*(W^{(L_j)} x_j) = 0$ for $j \leq i-1$, and $f_i^*(W^{(L_i)} x_i) = 1$. Define J_i as follows:

$$J_i v := J_{i-1} v + (y_i - J_{i-1} W^{(L_i)} x_i) \otimes f_i^*(v). \tag{6}$$

Thus, $J_i W^{(L_i)} x_i = J_{i-1} W^{(L_i)} x_i + (y_i - J_{i-1} W^{(L_i)} x_i) \cdot 1 = y_i$ and $J_i W^{(L_j)} x_j = J_{i-1} W^{(L_j)} x_j = y_j$, for $j \leq i-1$ since $f_i^*(W^{(L_j)} x_j) = 0$ and (5). By induction, for $1 \leq i \leq n_0$: $J_i W^{(L_j)} x_j = y_j$ for $j \leq i$. Finally, by setting $J := J_{n_0}$, we have that $Y_i = JW^{(L_i)} x_i$ for all $1 \leq i \leq n_0$. ■

Remark 5 *On J as an Isomorphism: Define the invertibility metric*

$$\delta := \max_{1 \leq i \leq N} \|Y_i - J_{i-1} W^{(L_i)} X_i\|.$$

By iterative construction, $J = I_d + \sum_{i=1}^N (Y_i - J_{i-1} W^{(L_i)} X_i) \otimes f_i^*$, so $\|J - I_d\| \leq N\delta \cdot \max_i \|f_i^*\|$. Thus, J is an isomorphism if $\|J - I_d\| < 1$ (see Lemma 2.1, p 192 (8)), achieved when $\delta < 1/(N \cdot \max_i \|f_i^*\|)$. The restriction $L_i \in \{0, \ldots, d-1\}$ in Theorem 4 follows from Lemma 3, as $W^{(L)} X$ cycles with period d up to scaling. Beyond this, higher L only rescales earlier terms, adding no new linear independence information.

Remark 6 (Blockwise feasibility of the recursive independence condition) *Condition (4) is a certificate for grouping anchors within a single local interpolant, not a global feasibility obstruction. Indeed, let $\Xi_{\text{sub}} = \{(x_i, y_i)\}_{i=1}^{n_0} \subset \mathbb{R}^d \times \mathbb{R}^C$ and assume only that $x_i \neq 0$ for all i. Then there always exists a partition $\Xi_{\text{sub}} = D_0 \sqcup \cdots \sqcup D_{K-1}$, $K \leq n_0$, such that condition (4) holds on every block: it suffices to take the singleton partition and choose $L_i = 0$, so that $W^{(L_i)} x_i = x_i \neq 0$. Thus exact interpolation is always feasible blockwise after a sufficiently fine partition. The role of condition (4) is therefore to control how many anchors can be compressed into the same block, i.e. the size of K, rather than feasibility in principle.*

Remark 7 *The recursive linear independence condition* (4) *is a sufficient condition for exact interpolation and serves as a clean theoretical certificate of realizability. In practice, however, we do not enforce it as a hard requirement in noisy or compressed regimes, and exact independence is not needed for stable performance. Phase 1 tests span membership via a least–squares residual, which we use as a practical proxy for independence, and Phase 2 employs regularized least–squares interpolation to handle rank deficiency and noise. This defines what we call the* relaxed (non–exact) interpolation *modality: exact fitting is intentionally sacrificed in favor of stability and robustness, and this is the regime used in all our main experiments.*

Remark 8 (Infinite-dimensional extension of exact interpolation) *The proof of Theorem 4 is not fundamentally finite-dimensional. Its core ingredient is the Hahn–Banach extension Theorem which applies equally in infinite-dimensional Banach spaces. More precisely, once one has vectors $W^{(L_i)}x_i$ satisfying the analogue of condition (4) in infinite dimensions, the same rank-one update construction used above produces, a linear operator J interpolating the prescribed targets. Thus the argument extends, with the same proof structure, to infinite-dimensional Banach spaces after replacing the wrapped finite-dimensional shift by a bounded classical weighted backward shift on a sequence Banach space as defined in (1), and allowing $L_i \in \mathbb{N}_0$ rather than $L_i \in \{0,\dots,d-1\}$. We do not develop the full infinite-dimensional theory here, but emphasize that the exact interpolation mechanism itself already persists in that setting.*

The main goal of this section is operator realizability, which is mediated by orbit dictionaries, and the first structural step is the construction of a data-induced orbit scaffold associated with the dataset Ξ.

Select a small anchor subset $\Xi_{\text{sub}} \subseteq \Xi$ and a partition $\{D_k\}_{k=1}^K$ of Ξ_{sub}. The construction of the scaffold proceeds as follows: we learn a wrapped weighted backward shift operator W, then we apply the exact interpolation Theorem 4 blockwise to the sets D_k, producing linear operators J_k that interpolate exactly on each block.

More precisely, for every training pair $(x_i, y_i) \in \Xi_{\text{sub}}$ with $x_i \in D_k$, , assume one can choose an integer $L_i \in \{0,\dots,d-1\}$ so that condition (4) holds on D_k; then Theorem 4 yields a linear operator J_k such that $y_i = J_k W^{(L_i)} x_i$. Thus orbit iterates generated by W are mapped by the corresponding operators J_k exactly onto the target outputs on the anchor subset. This leads to the orbit-based scaffold associated with Ξ_{sub}:

$$U(\Xi_{\text{sub}}) := \operatorname{span}\Big\{ J_k W^{(m)} x_i : (x_i, y_i) \in \Xi_{\text{sub}},\ 1 \le k \le K,\ 0 \le m \le d-1 \Big\} \subseteq \mathbb{R}^C.$$

Hence $U(\Xi_{\text{sub}})$ is a finite-rank, data-induced subspace of the output Hilbert space encoding the orbit geometry of the dataset. The following decomposition is an immediate consequence of the triangle inequality and is therefore purely structural. It identifies the two fundamental bottlenecks of our learning framework.

Theorem 9 (Two–term error decomposition for operator realization on an orbit scaffold) *Let $\Xi = \{(x_i, y_i)\}_{i=1}^n \subset \mathbb{R}^d \times \mathbb{R}^C$ be a finite dataset, $\Xi_{\text{sub}} \subseteq \Xi$, and let $A : \mathbb{R}^d \to \mathbb{R}^C$ be a linear operator. Let $U(\Xi_{\text{sub}})$ be the orbit scaffold relative to Ξ_{sub} and $\Pi_{U(\Xi_{\text{sub}})} : \mathbb{R}^C \to \mathbb{R}^C$ the associated orthogonal projector and $\hat{y} : \mathbb{R}^d \to \mathbb{R}^C$. Then the following decomposition holds:*

$$\|A - \hat{y}\|_\Xi \;\le\; \underbrace{\|A - \Pi_{U_{\Xi_{\text{sub}}}} A\|_\Xi}_{\varepsilon_{\text{proj}}(A, U_{\Xi_{\text{sub}}}, \Xi)} \;+\; \underbrace{\|\Pi_{U_{\Xi_{\text{sub}}}} A - \hat{y}\|_\Xi}_{\varepsilon_{\text{param}}(A, U_{\Xi_{\text{sub}}}, \Xi)}. \tag{7}$$

In practice, the projection term $\varepsilon_{\text{proj}}$ is typically the dominant bottleneck; the experiments in section 4 confirm that once a high-quality scaffold is found, the parametrization term collapses reliably with our constructive realizer mechanism CCDS introduced and developed below.

The meaning of the decomposition (7) is not limited to an upper bound for a particular realization algorithm. The dataset Ξ induces a scaffold $U(\Xi_{\text{sub}})$ (a geometric "DNA" extracted from Ξ) that is a data-only object, independent of the operator A. The projection term $\varepsilon_{\text{proj}}(A, U(\Xi_{\text{sub}}), \Xi)$ depends only on A and on this dataset-induced scaffold, and measures how much of the action of A on Ξ lies in the orbit geometry intrinsic to Ξ_{sub}. Equivalently, it quantifies how much of the structure inherent to the operator A is carried by the dataset Ξ_{sub}. Small values indicate that the outputs Ax are largely contained in the subspaces encoded by $U(\Xi_{\text{sub}})$, while large values signal directions of A that are transversal to the dataset's orbit structure. The parametrization term $\varepsilon_{\text{param}}(A, U(\Xi_{\text{sub}}), \Xi)$ instead measures whether, once this geometric compatibility holds, the realizer $\hat{y}$ constrained to

$U(\Xi_{\rm sub})$ can constructively synthesize the projected target. The decomposition therefore separates operator complexity, operator–dataset alignment, and algorithmic realizability, and can be read simultaneously as an approximation bound, a realizability statement, and a geometric diagnostic.

Theorem 9 identifies scaffold selection and realization as the two fundamental learning bottlenecks in orbit-based models. These two components will be the focus of the remainder of this section.

We begin addressing both bottlenecks within this paper. On the geometric side, we cast scaffold selection as an orbit-constrained optimization problem and develop a certifiable deviation-controlled algorithm for selecting the scaffold. On the realization side, we introduce a certifiable deviation-controlled algorithm for constructing a realizer on a fixed scaffold. When both the projection and parametrization errors collapse, the target linear operator can be constructively realized on the data subspace with arbitrarily small error. In particular, if the target operator is the data-dependent linear operator induced by attention on the input sequence (i.e., for a fixed token sequence $X = (x_1, \ldots, x_n) \in \mathbb{R}^{d\times n}$, single-head attention determines a data-dependent linear operator $A(X)$ acting on token/value representations, and the corresponding output is obtained by applying the map $\nu \mapsto A(X)\nu$), then attention requires no separate treatment: it is simply a concrete instance of the general operator-realization problem studied here. Attention therefore appears as an immediate special case of a broader operator-realization principle. Taken together, these observations suggest that attention may be viewed less as a primitive architectural ingredient and more as a problem of realizing input-dependent linear operators, a task that may be approached, for instance, through our orbit-based framework without relying on softmax algebra or probabilistic arguments.

We now introduce one phase for each bottleneck: Phase $1^{\rm real}$ selects the scaffold, and Phase $2^{\rm real}$ constructs the realizer $\hat{y}$.

Phase convention. In this section, Phase $1^{\rm real}$ and Phase $2^{\rm real}$ refer to the two stages of the realizability framework. Later, in the predictive part (section 3), we will use Phase $1^{\rm pred}$, Phase $2^{\rm pred}$, and Phase $3^{\rm pred}$ for the corresponding predictive pipeline.

2.1 Projection error and scaffold selection problem.

We first show that the existence of optimal scaffolds under which projection error collapses, is a mathematically well-posed problem. Denote by $\mathcal{U}$ the set of all scaffolds relative to Ξ.

Theorem 10 (Existence of the best orbit-constrained scaffold) *Let $\Xi = \{(x_i, y_i)\}_{i=1}^n \subset \mathbb{R}^d \times \mathbb{R}^C$, and let $A : \mathbb{R}^d \to \mathbb{R}^C$ be linear, then there exists $U_\star \in \mathcal{U}$ such that $\|A - \Pi_{U_\star} A\|_\Xi = \min_{U\in\mathcal{U}} \|A - \Pi_U A\|_\Xi$.*

Proof First, let us compute an upper bound for the dimension of any scaffold U relative to Ξ. Pick any subdataset $\Xi_{\rm sub} \subset \Xi$ and assume it is partitioned as $\Xi_{\rm sub} = D_0 \sqcup \cdots \sqcup D_{K-1}$. Then, each sample $x_i \in \Xi_{\rm sub}$ belongs to exactly one block D_k. Hence, in the dictionary construction, x_i contributes only the d atoms $J_k W^{(m)} x_i, \quad 0 \le m \le d-1$, associated with its own interpolant J_k. Therefore the total number of generating atoms is at most $d\,|\Xi_{\rm sub}|$, and since the scaffold is a subspace of $\mathbb{R}^C$, we obtain $\dim U \le \min\big(C,\ d\,|\Xi_{\rm sub}|\big) \le \min\big(C,\ d\,n\big)$. If $\mathcal{U}_k$ denotes the set of all k-dimensional scaffolds relative to Ξ. Hence,

$$\|A - \Pi_{U_\star} A\|_\Xi = \min_{1\le k\le \min(C,dn)} \min_{U\in\mathcal{U}_k} \|A - \Pi_U A\|_\Xi.$$

In order to conclude our proof, it will be enough to prove the existence of a best orbit-constrained scaffold of dimension k, for all $1 \le k \le \min(C, dn)$. Since Ξ is finite, there are only finitely many admissible choices of subsets and partitions on which the exact interpolation theorem can be applied. Each such admissible choice produces one scaffold. Therefore, for each fixed k, the family of all k-dimensional scaffolds induced by Ξ is finite, hence compact. On the other hand, the map $U \mapsto \Pi_U$ is continuous (see, e.g., (8)). Define $f(U) = \|A - \Pi_U A\|_\Xi$. Since A is fixed and $U \mapsto \Pi_U$ is continuous in operator norm, f is continuous on $\mathcal{U}_k$. Because $\mathcal{U}_k$ is compact and f is continuous, the extreme value theorem implies that f attains its minimum on $\mathcal{U}_k$. Hence, there exists $U_\star^{(k)} \in \mathcal{U}_k$ such that $f(U_\star^{(k)}) = \min_{U\in\mathcal{U}_k} f(U) = \min_{U\in\mathcal{U}_k} \|A - \Pi_U A\|_\Xi$. ∎

From existence to computation. Theorem 10 guarantees the existence of an optimal orbit-constrained scaffold, but it is nonconstructive. The missing issue is algorithmic: given the admissible orbit geometry induced by the dataset, how can one enlarge the scaffold in a controlled way so as to decrease the projection term. This is not a classical unconstrained low-rank approximation problem, because the admissible subspaces are not arbitrary subspaces of $\mathbb{R}^C$; they must arise from the orbit mechanism described above. We therefore isolate the projection term and develop a constructive *scaffold-level* deviation-control theory. The object being controlled is the projection residual associated with nested admissible scaffold expansions.

The resulting theory has four parts. First, the projection error is rewritten as a Frobenius residual of a snapshot matrix. Second, for one-dimensional scaffold expansions, we prove an exact energy identity describing the drop in residual energy. Third, we characterize all one-step feasible drops and show how to realize any prescribed feasible drop explicitly. Fourth, we obtain a constructive theorem showing that any stepwise feasible projection-error profile can be enforced exactly. The greedy spectral step then appears as a distinguished special case.

Let $\Xi = \{x_i\}_{i=1}^N \subset \mathbb{R}^d$, and let $A : \mathbb{R}^d \to \mathbb{R}^C$ be linear. Define the squared dataset seminorm by $\|A\|_{\Xi,2}^2 := \frac{1}{N}\sum_{i=1}^N \|Ax_i\|_2^2$. Fix an admissible scaffold space $V \subset \mathbb{R}^C$ arising from the orbit construction, and let $U \subseteq V$ be an admissible scaffold.

Proposition 11 (Snapshot representation of the projection error) *Define the snapshot matrix $Y := \begin{bmatrix} Ax_1, \dots, Ax_N \end{bmatrix} \in \mathbb{R}^{C\times N}$. Then, for every orthogonal projector Π_U,*

$$\|A - \Pi_U A\|_{\Xi,2}^2 = \frac{1}{N}\|(I - \Pi_U)Y\|_F^2.$$

Proof By definition of the squared dataset seminorm,

$$\|A - \Pi_U A\|_{\Xi,2}^2 = \frac{1}{N}\sum_{i=1}^N \|(A - \Pi_U A)x_i\|_2^2 = \frac{1}{N}\sum_{i=1}^N \|(I - \Pi_U)Ax_i\|_2^2.$$

On the other hand, the i-th column of $(I - \Pi_U)Y$ is precisely $(I - \Pi_U)Ax_i$. Since the Frobenius norm is the sum of the squared Euclidean norms of the columns, we obtain $\|(I - \Pi_U)Y\|_F^2 = \sum_{i=1}^N \|(I - \Pi_U)Ax_i\|_2^2$. Dividing by N gives the claim. ∎

One-step scaffold expansion. We now work in the ambient Hilbert space $H = \mathbb{R}^C$ with its standard inner product. Let $U_0 \subset U_1 \subset \cdots \subset U_T \subset V$ be a nested chain of admissible subspaces satisfying $\dim(U_t/U_{t-1}) = 1,\ t = 1, \dots, T$. Equivalently, for each t there exists a unit vector $u_t \in V \cap U_{t-1}^\perp$ such that $U_t = U_{t-1} \oplus \operatorname{span}\{u_t\}$. For each t, define the residual matrix and its energy by

$$R_t := (I - \Pi_{U_t})Y, \qquad E_t := \|R_t\|_F^2.$$

Lemma 12 (One-step energy identity) *Let $U_t = U_{t-1} \oplus \operatorname{span}\{u\},\ u \perp U_{t-1}$, and $\|u\| = 1$. Then $E_t = E_{t-1} - \|u^\top Y\|_2^2 = E_{t-1} - \|u^\top R_{t-1}\|_2^2$.*

Proof Since $u \perp U_{t-1}$ and $\|u\| = 1$, the orthogonal projector onto U_t is $\Pi_{U_t} = \Pi_{U_{t-1}} + uu^\top$. Therefore

$$R_t = (I - \Pi_{U_t})Y = (I - \Pi_{U_{t-1}} - uu^\top)Y = R_{t-1} - u(u^\top Y).$$

Taking Frobenius norms and using $\|A - B\|_F^2 = \|A\|_F^2 - 2\langle A, B\rangle_F + \|B\|_F^2$, we get

$$\|R_t\|_F^2 = \|R_{t-1}\|_F^2 - 2\langle R_{t-1}, u(u^\top Y)\rangle_F + \|u(u^\top Y)\|_F^2.$$

We compute the two remaining terms separately.

First, using the Frobenius inner product $\langle A, B\rangle_F = \mathrm{tr}(A^\top B)$,

$$\begin{aligned}
\langle R_{t-1}, u(u^\top Y)\rangle_F &= \mathrm{tr}\left(R_{t-1}^\top u(u^\top Y)\right) \\
&= u^\top Y R_{t-1}^\top u \\
&= u^\top R_{t-1} R_{t-1}^\top u \qquad \text{because } u^\top R_{t-1} = u^\top Y \\
&= \|u^\top R_{t-1}\|_2^2.
\end{aligned}$$

The identity $u^\top R_{t-1} = u^\top Y$ follows from $R_{t-1} = (I - \Pi_{U_{t-1}})Y$ and the fact that $u \perp U_{t-1}$, which implies $u^\top \Pi_{U_{t-1}} = 0$. Second, since $u(u^\top Y)$ is rank one and $\|u\| = 1$,

$$\|u(u^\top Y)\|_F^2 = \|u\|_2^2 \, \|u^\top Y\|_2^2 = \|u^\top Y\|_2^2.$$

But again $u^\top R_{t-1} = u^\top Y$, so $\|u(u^\top Y)\|_F^2 = \|u^\top R_{t-1}\|_2^2$. Substituting into the Frobenius expansion yields

$$E_t = E_{t-1} - 2\|u^\top R_{t-1}\|_2^2 + \|u^\top R_{t-1}\|_2^2 = E_{t-1} - \|u^\top R_{t-1}\|_2^2.$$

Since $u^\top R_{t-1} = u^\top Y$, the first identity follows as well. ∎

Feasible one-step drops. Fix an admissible scaffold $U \subseteq V$, and write

$$S := V \cap U^\perp, \qquad R := (I - \Pi_U)Y, \qquad E := \|R\|_F^2.$$

Thus S is the admissible complement in which a new one-dimensional scaffold increment must lie. For $u \in S$ with $\|u\| = 1$, define the one-step capture functional $q(u) := \|u^\top R\|_2^2$. This is the amount of residual energy removed by adjoining the direction u.

Lemma 13 (Feasible one-step residual energies) *Let $M := \Pi_S R R^\top \Pi_S$. Then M is a symmetric positive semidefinite operator with range contained in S. Define*

$$\lambda_{\min} := \min_{\substack{u \in S \\ \|u\|=1}} u^\top M u, \qquad \lambda_{\max} := \max_{\substack{u \in S \\ \|u\|=1}} u^\top M u.$$

Then the following hold:

1. *$q(u) = u^\top M u$ for every $u \in S$ with $\|u\| = 1$.*
2. *The set of achievable capture energies $\{q(u) : u \in S,\ \|u\| = 1\}$ is exactly the interval $[\lambda_{\min}, \lambda_{\max}]$.*
3. *Consequently, the set of achievable next residual energies is exactly $\big[E - \lambda_{\max},\, E - \lambda_{\min}\big]$.*

Proof We first note that M is symmetric because both Π_S and $RR^\top$ are symmetric: $M^\top = (\Pi_S R R^\top \Pi_S)^\top = \Pi_S R R^\top \Pi_S = M$. Moreover, for every $x \in \mathbb{R}^C$, $x^\top M x = x^\top \Pi_S R R^\top \Pi_S x = \|R^\top \Pi_S x\|_2^2 \geq 0$, so M is positive semidefinite.

Proof of (1). Let $u \in S$ with $\|u\| = 1$. Since $\Pi_S u = u$, we have

$$u^\top M u = u^\top \Pi_S R R^\top \Pi_S u = u^\top R R^\top u.$$

On the other hand,

$$u^\top RR^\top u = (u^\top R)(u^\top R)^\top = \|u^\top R\|_2^2 = q(u).$$

Thus $q(u) = u^\top Mu$.

Proof of (2). Because $u \mapsto u^\top Mu$ is continuous on the unit sphere of S, the extreme value theorem shows that the minimum and maximum are attained. Hence

$$\{q(u) : u \in S,\ \|u\| = 1\} \subseteq [\lambda_{\min}, \lambda_{\max}],$$

and both endpoints belong to the image. It remains to show that every intermediate value is also attained. If $\dim S = 1$, then the unit sphere of S is $\{u, -u\}$ for some unit vector u, and since the quadratic form is even, $q(-u) = q(u)$, the image is a singleton. In this case $\lambda_{\min} = \lambda_{\max}$, so the image is precisely the degenerate interval $[\lambda_{\min}, \lambda_{\max}]$.

Assume now that $\dim S \geq 2$. Then the unit sphere of S is connected. Since $u \mapsto u^\top Mu$ is continuous, its image is a connected subset of $\mathbb{R}$, hence an interval. Because this interval contains both $\lambda_{\min}$ and $\lambda_{\max}$, it must contain the whole closed interval between them. Therefore

$$\{q(u) : u \in S,\ \|u\| = 1\} = [\lambda_{\min}, \lambda_{\max}].$$

Proof of (3). By Lemma 12, after adjoining a unit direction $u \in S$ the new residual energy is $E^+ = E - q(u)$. Since $q(u)$ ranges exactly over $[\lambda_{\min}, \lambda_{\max}]$, the possible values of E^+ are exactly $E - [\lambda_{\min}, \lambda_{\max}] = [E - \lambda_{\max},\ E - \lambda_{\min}]$. This proves the claim. ■

Lemma 14 (Explicit two-dimensional construction) *Assume $\dim S \geq 2$, and let $v_1, v_2 \in S$ be orthonormal eigenvectors of M with eigenvalues $\lambda_1 \geq \lambda_2$. Then for every target $e \in [\lambda_2, \lambda_1]$, there exists a unit vector $u(\theta) := \cos\theta\, v_1 + \sin\theta\, v_2 \in S$ such that $q(u(\theta)) = e$. More precisely, if $\lambda_1 > \lambda_2$, one may choose θ so that $\cos^2\theta = \frac{e-\lambda_2}{\lambda_1-\lambda_2}$. In that case, adjoining $u(\theta)$ enforces the one-step update $E^+ = E - e$.*

Proof Let $u(\theta) = \cos\theta\, v_1 + \sin\theta\, v_2$. Because $v_1, v_2 \in S$, we have $u(\theta) \in S$. Since v_1 and v_2 are orthonormal,

$$\|u(\theta)\|_2^2 = \cos^2\theta\, \|v_1\|_2^2 + \sin^2\theta\, \|v_2\|_2^2 + 2\cos\theta\sin\theta\, \langle v_1, v_2\rangle = \cos^2\theta + \sin^2\theta = 1.$$

Thus $u(\theta)$ is admissible. By Lemma 13, $q(u(\theta)) = u(\theta)^\top Mu(\theta)$. Expanding and using the eigenvector relations $Mv_j = \lambda_j v_j$, we get

$$\begin{aligned}
q(u(\theta)) &= (\cos\theta\, v_1 + \sin\theta\, v_2)^\top M(\cos\theta\, v_1 + \sin\theta\, v_2) \\
&= \cos^2\theta\, v_1^\top Mv_1 + \sin^2\theta\, v_2^\top Mv_2 + \cos\theta\sin\theta\, v_1^\top Mv_2 + \cos\theta\sin\theta\, v_2^\top Mv_1 \\
&= \cos^2\theta\, \lambda_1 + \sin^2\theta\, \lambda_2,
\end{aligned}$$

because $v_1^\top Mv_2 = \lambda_2\langle v_1, v_2\rangle = 0$ and similarly $v_2^\top Mv_1 = 0$.

If $\lambda_1 = \lambda_2$, then the interval $[\lambda_2, \lambda_1]$ is a single point and $q(u(\theta)) = \lambda_1 = \lambda_2$ for every θ, so the claim is immediate. Assume therefore that $\lambda_1 > \lambda_2$. Since

$$q(u(\theta)) = \cos^2\theta\, \lambda_1 + \sin^2\theta\, \lambda_2 = \lambda_2 + \cos^2\theta(\lambda_1 - \lambda_2),$$

the condition $q(u(\theta)) = e$ is equivalent to $\cos^2\theta = \frac{e-\lambda_2}{\lambda_1-\lambda_2}$. Because $e \in [\lambda_2, \lambda_1]$, the right-hand side belongs to $[0, 1]$, so such a θ exists. For this choice of θ, we obtain $q(u(\theta)) = e$. Finally, by Lemma 12, adjoining $u(\theta)$ changes the energy by $E^+ = E - q(u(\theta)) = E - e$. ■

Theorem 15 (Constructive deviation-control principle for scaffold selection) *Let $V \subset \mathbb{R}^C$ be a closed admissible scaffold space. Let $Y \in \mathbb{R}^{C\times N}$ be fixed, and define*

$$R_0 := Y, \qquad U_0 := \{0\}, \qquad E_0 := \|R_0\|_F^2.$$

Fix an integer $T \geq 1$ and a target energy profile $E_0 \geq E_1 \geq \cdots \geq E_T \geq 0$. For each $t = 1, \ldots, T$, define the admissible complement $S_{t-1} := V \cap U_{t-1}^{\perp}$ and the compressed residual covariance operator

$$M_{t-1} := \Pi_{S_{t-1}} R_{t-1} R_{t-1}^{\top} \Pi_{S_{t-1}}.$$

Let

$$\lambda_{\min}^{(t-1)} := \min_{\substack{u\in S_{t-1}\\ \|u\|=1}} u^{\top} M_{t-1} u, \qquad \lambda_{\max}^{(t-1)} := \max_{\substack{u\in S_{t-1}\\ \|u\|=1}} u^{\top} M_{t-1} u.$$

Assume the feasibility inequalities

$$E_{t-1} - E_t \in \big[\lambda_{\min}^{(t-1)}, \lambda_{\max}^{(t-1)}\big], \qquad t = 1, \ldots, T.$$

Then there exists a deterministic sequence of unit vectors $u_t \in S_{t-1}$ and a nested scaffold chain $U_t := U_{t-1} \oplus \operatorname{span}\{u_t\} \subset V$ such that, with $R_t := (I - \Pi_{U_t})Y$, one has

$$\|R_t\|_F^2 = E_t, \qquad t = 1, \ldots, T.$$

Moreover, whenever $\dim S_{t-1} \geq 2$, one may choose u_t explicitly by Lemma 14 so as to realize the prescribed drop $E_{t-1} - E_t$ exactly.

Proof We argue by induction on t. For $t = 0$, the conclusion holds by definition:

$$U_0 = \{0\}, \qquad R_0 = Y, \qquad \|R_0\|_F^2 = E_0.$$

Assume now that for some $t \in \{1, \ldots, T\}$ we have already constructed $U_{t-1} \subset V$ and

$$R_{t-1} = (I - \Pi_{U_{t-1}})Y$$

in such a way that $\|R_{t-1}\|_F^2 = E_{t-1}$. At step t, the admissible unit directions are precisely the unit vectors in $S_{t-1} = V \cap U_{t-1}^{\perp}$. By Lemma 13, the set of achievable one-step capture energies

$$q(u) = \|u^{\top} R_{t-1}\|_2^2, \qquad u \in S_{t-1},\ \|u\| = 1,$$

is exactly the interval $\big[\lambda_{\min}^{(t-1)}, \lambda_{\max}^{(t-1)}\big]$. By the feasibility assumption, $E_{t-1} - E_t \in \big[\lambda_{\min}^{(t-1)}, \lambda_{\max}^{(t-1)}\big]$. Hence there exists a unit vector $u_t \in S_{t-1}$ such that $q(u_t) = E_{t-1} - E_t$. Define $U_t := U_{t-1} \oplus \operatorname{span}\{u_t\}$. Since $u_t \in V$, we have $U_t \subset V$, and since $u_t \perp U_{t-1}$, the sum is orthogonal and therefore $U_{t-1} \subset U_t$, so the chain remains nested.

Now define $R_t := (I - \Pi_{U_t})Y$. Applying Lemma 12, we obtain

$$\|R_t\|_F^2 = \|R_{t-1}\|_F^2 - q(u_t) = E_{t-1} - (E_{t-1} - E_t) = E_t.$$

This completes the induction step. Therefore, by induction, the whole sequence

$$U_0 \subset U_1 \subset \cdots \subset U_T \subset V$$

can be constructed so that $\|R_t\|_F^2 = E_t$ for every $t = 1, \ldots, T$.

For the final assertion, assume $\dim S_{t-1} \geq 2$. Then, instead of choosing u_t abstractly from Lemma 13, one may choose it explicitly by Lemma 14, using two orthonormal eigenvectors of M_{t-1} and selecting the mixing angle so that the capture energy equals the prescribed drop $E_{t-1} - E_t$. ∎

Corollary 16 (Projection-error profile in dataset norm) *For the scaffold chain constructed in Theorem 15, one has*

$$\varepsilon_{\mathrm{proj}}(U_t)^2 = \|A_r - \Pi_{U_t} A_r\|_{\Xi,2}^2 = \frac{1}{N}\|R_t\|_F^2 = \frac{E_t}{N}.$$

In particular, the theorem enforces a prescribed projection-error profile exactly in the squared dataset seminorm.

Proof By Proposition 11,

$$\|A_r - \Pi_{U_t} A_r\|_{\Xi,2}^2 = \frac{1}{N}\|(I - \Pi_{U_t})Y\|_F^2 = \frac{1}{N}\|R_t\|_F^2.$$

By Theorem 15, $\|R_t\|_F^2 = E_t$. Substituting this identity into the previous equality yields $\|A_r - \Pi_{U_t} A_r\|_{\Xi,2}^2 = \frac{E_t}{N}$, which is exactly the desired statement. ■

Corollary 17 (One-step infeasibility outside the scaffold feasibility interval) *In the setting of Lemma 13, let $e \notin [\lambda_{\min}, \lambda_{\max}]$. Then there exists no unit vector $u \in S$ such that $q(u) = e$. Equivalently, there is no one-dimensional admissible scaffold expansion $U^+ = U \oplus \mathrm{span}\{u\}$ whose residual energy satisfies $E^+ = E - e$. Consequently, any prescribed profile violating the stepwise interval constraints of Theorem 15 is unrealizable in the fixed admissible scaffold space.*

Proof By Lemma 13, $\{q(u) : u \in S,\ \|u\| = 1\} = [\lambda_{\min}, \lambda_{\max}]$. Hence if $e \notin [\lambda_{\min}, \lambda_{\max}]$, no admissible unit vector $u \in S$ can satisfy $q(u) = e$. By Lemma 12, this is equivalent to the nonexistence of a one-dimensional admissible scaffold expansion producing the residual-energy update $E^+ = E - e$. The final statement follows immediately by applying this argument at the first step where the prescribed profile violates the interval constraint. ■

Greedy maximal collapse as a distinguished special case. The constructive theorem above governs arbitrary *feasible* projection-error profiles. If one does not wish to prescribe the next drop in advance, the most natural canonical choice is instead to maximize the immediate decrease in residual energy.

Proposition 18 (Greedy one-step maximal collapse) *At step t, among all unit vectors $u \in S_{t-1}$, the choice maximizing the one-step drop $q(u) = \|u^\top R_{t-1}\|_2^2$ is any unit eigenvector of M_{t-1} associated with the largest eigenvalue $\lambda_{\max}^{(t-1)}$. The achieved drop is exactly $\lambda_{\max}^{(t-1)}$. Equivalently, such a maximizing direction is the left singular vector associated with the largest singular value of $\Pi_{S_{t-1}} R_{t-1}$.*

Proof By Lemma 13, for every unit vector $u \in S_{t-1}$, we have $q(u) = u^\top M_{t-1} u$. Therefore maximizing the one-step drop is equivalent to solving

$$\max_{\substack{u \in S_{t-1} \\ \|u\|=1}} u^\top M_{t-1} u.$$

Since M_{t-1} is symmetric, the Rayleigh–Ritz principle implies that this maximum is equal to the largest eigenvalue $\lambda_{\max}^{(t-1)}$, and that the maximizers are exactly the associated unit eigenvectors. Hence the largest achievable one-step drop is $\lambda_{\max}^{(t-1)}$. For the singular-vector formulation, set $B := \Pi_{S_{t-1}} R_{t-1}$. Then

$$BB^\top = (\Pi_{S_{t-1}} R_{t-1})(\Pi_{S_{t-1}} R_{t-1})^\top = \Pi_{S_{t-1}} R_{t-1} R_{t-1}^\top \Pi_{S_{t-1}} = M_{t-1},$$

because $\Pi_{S_{t-1}}$ is symmetric. The eigenvectors of $BB^\top$ are precisely the left singular vectors of B, and the corresponding eigenvalues are the squares of the singular values. Therefore the maximizing eigenvectors of M_{t-1} are exactly the left singular vectors associated with the largest singular value of $\Pi_{S_{t-1}} R_{t-1}$. ■

Transition to realizability and prediction. The scaffold-selection mechanism developed above addresses only the projection term in the two-term decomposition. Once an admissible scaffold has been selected and its projection error has been reduced in a certified way, the remaining problem is constructive synthesis on that scaffold, namely the collapse of the parametrization term. We now turn to this second bottleneck (parametrization term).

In the predictive setting (developed later in Section 3), the same structural logic persists: the scaffold is first selected geometrically, the projected target is constructively realized on that scaffold, and prediction is then reduced to learning a coefficient law that extends this constructive realization over the associated orbit dictionary.

2.2 Collapsing the parametrization error.

Once a scaffold has been fixed and the projection error $\varepsilon_{\text{proj}}$ controlled, the remaining approximation error lies in the realization of the compressed operator on that scaffold. The following example shows that exact span containment does not guarantee stable or efficient collapse of the parametrization error. In particular, a purely greedy innovation rule is insufficient: it has no mechanism to enforce a prescribed decrease of the residual norm, nor to prevent oscillatory selection among nearly collinear atoms. The realizability problem is therefore not solved by classical greedy pursuit alone, even when the projection error is zero. This pathology motivates deviation-controlled solvers such as our CCDS introduced here (all details below), which explicitly regulate the residual trajectory and stabilize the coefficient growth, ensuring constructive convergence even in the presence of severe geometric degeneracies of the scaffold.

Example 3 (Canonical ill–conditioned realizability example) *Consider $\mathbb{R}^2$ with scaffold atoms $v_1 = (1,0)^\top$, $v_2 = (\cos\theta, \sin\theta)^\top$, with $0 < \theta \ll 1$, and target $y = (0,1)^\top$. Since $\operatorname{span}\{v_1, v_2\} = \mathbb{R}^2$, the projection error vanishes and exact realization is possible: $\alpha_1 v_1 + \alpha_2 v_2 = y$. The unique solution is $\alpha_2 = 1/\sin\theta$ and $\alpha_1 = -\cos\theta/\sin\theta$, so $|\alpha_1| + |\alpha_2| \sim \theta^{-1} \to \infty$ as $\theta \to 0$. Thus exact realizability requires arbitrarily large coefficients with near–perfect cancellation.*

Moreover, since the angle between v_1 and v_2 is θ, successive orthogonal projections alternate between the two atoms and the residual admits the exact closed form $\|r_t\| = \cos^t\theta$. In particular, for small θ one has $\cos\theta = 1 - \theta^2/2 + O(\theta^4)$, so $\|r_t\| = \cos^t\theta \approx \exp(-t\theta^2/2)$; hence a constant–factor reduction of the residual requires on the order of $1/\theta^2$ alternating steps as $\theta \to 0$. The corresponding coefficients satisfy $a_{2k+1} = \cos^{2k}\theta \sin\theta$ and $a_{2k+2} = -\cos^{2k+1}\theta \sin\theta$: they alternate in sign and are successively scaled by factors of $\cos\theta$, building up large cancelling contributions whose sum converges to the ill–conditioned exact coefficients $\alpha_1 = -\cos\theta/\sin\theta$ and $\alpha_2 = 1/\sin\theta$. This explicit geometric–series accumulation makes precise that exact realizability is achieved only through the delicate cancellation of two large, nearly equal contributions, rendering the parametrization extremely sensitive and explaining the observed coefficient blow-up in the near-collinear regime.

Geometrically, the atoms become almost collinear. Indeed, one has $\langle r_{2k}, v_1\rangle = 0$ and $\langle r_{2k}, v_2\rangle = a_{2k+1} = \cos^{2k}\theta \sin\theta$, hence $\langle r_{2k}, v_1\rangle - \langle r_{2k}, v_2\rangle = -a_{2k+1}$. Likewise, $\langle r_{2k+1}, v_2\rangle = 0$ and $\langle r_{2k+1}, v_1\rangle = a_{2k+2} = -\cos^{2k+1}\theta \sin\theta$, hence $\langle r_{2k+1}, v_1\rangle - \langle r_{2k+1}, v_2\rangle = a_{2k+2}$. Therefore the atoms are selected in strict alternation, and the correlation gap has magnitude $|a_{t+1}| = \cos^t\theta \sin\theta = O(\sin(\theta))$, so the selection rule becomes arbitrarily sensitive to perturbations as $\theta \to 0$.

Example 3 shows that even with zero projection error, classical greedy pursuit is insufficient to guarantee stable collapse of parametrization error. Controlled innovation is required to prevent oscillation and coefficient blow–up, which motivates our deviation–controlled solvers introduced below.

CCDS: Constructive Deviation-Controlled Synthesis for Phase 3^{real}. Let $(\mathcal{H}, \langle\cdot,\cdot\rangle)$ be a real Hilbert space with norm $\|x\| = \sqrt{\langle x, x\rangle}$. For any closed subspace $Y \subset \mathcal{H}$ let Π_Y denote the orthogonal projector and

$$\rho(x, Y) \;:=\; \inf_{y\in Y} \|x - y\| \;=\; \|x - \Pi_Y x\|$$

the distance from x to Y.

Nested scaffolds. Fix a nested chain of closed subspaces $Y_0 \subset Y_1 \subset \cdots \subset Y_T \subset \mathcal{H}$ with $\dim(Y_t/Y_{t-1}) = 1$ for $t = 1, \ldots, T$.[1] For each t pick a unit *innovation direction* $u_t \in Y_t \cap Y_{t-1}^{\perp}$ (hence $Y_t = Y_{t-1} \oplus \operatorname{span}\{u_t\}$).

Deviation profile. Given a target $y \in \mathcal{H}$, CCDS will construct an approximation $\hat{y} \in Y_T$ via a sequence $\hat{y}_0, \hat{y}_1, \ldots, \hat{y}_T$ with $\hat{y}_{t-1} \in Y_{t-1}$ and residuals $r_{t-1} := y - \hat{y}_{t-1}$. The key quantity controlled at step t is the *deviation of the updated residual from the* previous *scaffold*:

$$d_t \;:=\; \rho(r_t, Y_{t-1}) \qquad (t = 1, \ldots, T), \tag{8}$$

where $r_t := y - \hat{y}_t$ after the step-t update. The idea is: at each refinement step we add a new direction, so we can prescribe (within feasibility) how far the new residual remains from the preceding scaffold.

Scalar distance equation. Fix t and abbreviate $Y := Y_{t-1}$, $u := u_t \in Y^{\perp}$ and $r := r_{t-1}$. Consider the scalar function $\phi(\lambda) \;:=\; \rho(r - \lambda u,\ Y) \;=\; \|(I - \Pi_Y)(r - \lambda u)\| \;=\; \|a - \lambda u\|$, $\quad a := (I - \Pi_Y) r \in Y^{\perp}$. This is continuous, strictly convex in λ (in squared form), and its level sets can be solved exactly.

Lemma 19 (Distance equation: explicit solutions and feasibility) *Let $Y \subset \mathcal{H}$ be closed, let $u \in Y^{\perp}$ with $\|u\| = 1$, and let $r \in \mathcal{H}$. Set $a := (I - \Pi_Y) r \in Y^{\perp}$ and $\alpha := \langle a, u\rangle$. Then for any target $d \geq 0$, the equation*

$$\rho(r - \lambda u,\ Y) \;=\; d \tag{9}$$

has a real solution $\lambda \in \mathbb{R}$ if and only if

$$d \;\geq\; d_{\min} \;:=\; \inf_{\lambda\in\mathbb{R}} \rho(r - \lambda u, Y) \;=\; \sqrt{\|a\|^2 - \alpha^2}. \tag{10}$$

When feasible, (9) *has exactly two solutions*

$$\lambda_{\pm} \;=\; \alpha \pm \sqrt{d^2 - d_{\min}^2}. \tag{11}$$

Moreover, the unique minimizer is $\lambda^{\star} = \alpha$, attaining $\rho(r - \lambda^{\star} u, Y) = d_{\min}$.

Proof Since $u \in Y^{\perp}$ and $\|u\| = 1$,

$$\begin{aligned} \rho(r - \lambda u, Y)^2 &= \|a - \lambda u\|^2 \\ &= \|a\|^2 - 2\lambda\langle a, u\rangle + \lambda^2 \\ &= (\lambda - \alpha)^2 + (\|a\|^2 - \alpha^2). \end{aligned}$$

Thus $\rho(r - \lambda u, Y) = d$ is equivalent to $(\lambda - \alpha)^2 = d^2 - (\|a\|^2 - \alpha^2)$. Real solutions exist iff $d^2 \geq \|a\|^2 - \alpha^2$, i.e. $d \geq d_{\min}$, yielding (11). The minimizer is the vertex $\lambda^{\star} = \alpha$ with minimal value $d_{\min}$. ■

1. In applications, one either enforces $\dim(Y_t/Y_{t-1}) = 1$ by construction (e.g. one atom per step), or one works with a block version. The 1-dimensional increment case is the canonical CCDS primitive.

Constructive principle (finite-horizon CCDS). Lemma 19 yields a fully constructive, stepwise principle:

Theorem 20 (Constructive deviation-control principle) *Fix a nested chain $Y_0 \subset \cdots \subset Y_T$ with $\dim(Y_t/Y_{t-1}) = 1$ and innovation directions $u_t \in Y_t \cap Y_{t-1}^{\perp}$ with $\|u_t\| = 1$. Let $y \in \mathcal{H}$ and initialize $\hat{y}_0 = 0 \in Y_0$, $r_0 = y$. Fix any target deviation profile $(d_t)_{t=1}^T$ satisfying, for each t,*

$$d_t \;\geq\; d_{t,\min} \;:=\; \sqrt{\|a_t\|^2 - \alpha_t^2}, \quad a_t := (I - \Pi_{Y_{t-1}})r_{t-1},$$
$$\alpha_t := \langle a_t, u_t\rangle. \tag{12}$$

Then, there exists a (deterministic) sequence of coefficients $(\lambda_t)_{t=1}^T$ such that, defining

$$\hat{y}_t := \hat{y}_{t-1} + \lambda_t u_t \in Y_t, \qquad r_t := y - \hat{y}_t,$$

one has for every $t = 1, \ldots, T$ the exact deviation constraints $\rho(r_t, Y_{t-1}) = d_t$. The coefficients admit the explicit formula $\lambda_t = \alpha_t \pm \sqrt{d_t^2 - d_{t,\min}^2}$, with the sign chosen by any deterministic rule.

Proof Fix t. By construction $u_t \in Y_{t-1}^{\perp}$, and r_{t-1} is known from previous steps. The feasibility condition (12) is exactly (10) in Lemma 19 with $Y = Y_{t-1}$, $u = u_t$, $r = r_{t-1}$ and target $d = d_t$. Hence there exists λ_t solving $\rho(r_{t-1} - \lambda_t u_t, Y_{t-1}) = d_t$. Setting $\hat{y}_t := \hat{y}_{t-1} + \lambda_t u_t$ gives $r_t = y - \hat{y}_t = r_{t-1} - \lambda_t u_t$ and thus $\rho(r_t, Y_{t-1}) = d_t$. Proceed sequentially for $t = 1, \ldots, T$. ■

CCDS algorithm. We now define CCDS as an algorithmic Phase-3 solver that instantiates the constructive principle above on a chain induced from a dictionary.

Dictionary and ordering. Let $\mathcal{D} \subset \mathcal{H}$ be a unit-norm dictionary. Fix a deterministic ordering rule that, given a state (Y_{t-1}, r_{t-1}), selects a new atom $q_t \in \mathcal{D}$ such that $q_t \notin Y_{t-1}$. Define

$$u_t \;:=\; \frac{q_t - \Pi_{Y_{t-1}} q_t}{\|q_t - \Pi_{Y_{t-1}} q_t\|} \in Y_{t-1}^{\perp}, \quad Y_t := Y_{t-1} \oplus \operatorname{span}\{u_t\}. \tag{13}$$

Let $\eta_t := \|q_t - \Pi_{Y_{t-1}} q_t\| > 0$, and let $e_t \in \mathbb{R}^t$ denote the t-th canonical basis vector. Define recursively coefficient vectors $c^{(t)} \in \mathbb{R}^t$ by $c^{(1)} := (1)$, and, for $t \geq 2$,

$$c^{(t)} := \frac{1}{\eta_t}\left(e_t - \sum_{s=1}^{t-1} \langle q_t, u_s\rangle\, \widetilde{c}^{(s,t)}\right), \tag{14}$$

where $\widetilde{c}^{(s,t)} \in \mathbb{R}^t$ is the zero-padded extension of $c^{(s)}$, namely $\widetilde{c}^{(s,t)} := (c_1^{(s)}, \ldots, c_s^{(s)}, 0, \ldots, 0)$. Then set

$$\beta^{(t)} := (\beta^{(t-1)}, 0) + \lambda_t c^{(t)} \in \mathbb{R}^t. \tag{15}$$

Proposition 21 (Equivalent selected-atom representation of CCDS output) *For each $t = 1, \ldots, T$ one has $u_t = \sum_{j=1}^t c_j^{(t)} q_j$, and consequently $\widehat{y}_T = \sum_{t=1}^T \lambda_t u_t = \sum_{j=1}^T \beta_j^{(T)} q_j$.*

Proof We argue by induction on t. For $t = 1$, since $Y_0 = \{0\}$, one has $u_1 = q_1$, so $c^{(1)} = (1)$ and the claim holds. Assume now that for each $s < t$, $u_s = \sum_{j=1}^s c_j^{(s)} q_j$. Since $\{u_s\}_{s=1}^{t-1}$ is an orthonormal basis of Y_{t-1}, we have $\Pi_{Y_{t-1}} q_t = \sum_{s=1}^{t-1} \langle q_t, u_s\rangle u_s$. Substituting the inductive representation of each u_s,

$$q_t - \Pi_{Y_{t-1}} q_t = q_t - \sum_{s=1}^{t-1} \langle q_t, u_s\rangle \sum_{j=1}^{s} c_j^{(s)} q_j.$$

After zero-padding the vectors $c^{(s)}$ to $\mathbb{R}^t$, this becomes

$$q_t - \Pi_{Y_{t-1}} q_t = \sum_{j=1}^{t} \left(e_t - \sum_{s=1}^{t-1} \langle q_t, u_s \rangle \widetilde{c}^{(s,t)} \right)_j q_j.$$

Dividing by $\eta_t = \|q_t - \Pi_{Y_{t-1}} q_t\|$ yields $u_t = \sum_{j=1}^{t} c_j^{(t)} q_j$. This proves the first claim. Finally,

$$\widehat{y}_T = \sum_{t=1}^{T} \lambda_t u_t = \sum_{t=1}^{T} \lambda_t \sum_{j=1}^{t} c_j^{(t)} q_j = \sum_{j=1}^{T} \beta_j^{(T)} q_j,$$

by the recursive definition of $\beta^{(t)}$. This proves the result. ■

Definition 22 (CCDS algorithm with explicit atom-dictionary output) *Fix a horizon T and a deviation profile rule that provides targets $d_t \geq 0$ at each step. Given an input target $y \in H$, define $\widehat{y}_0 := 0$, $Y_0 := \{0\}$, $r_0 := y$, and initialize the dictionary-coefficient vector by $\beta^{(0)} := 0 \in \mathbb{R}^0$. For $t = 1, \ldots, T$:*

1. *Select $q_t \in \mathcal{D}$ deterministically and form u_t, Y_t by* (13).
2. *Compute $a_t := (I - \Pi_{Y_{t-1}}) r_{t-1}$, and $\alpha_t := \langle a_t, u_t \rangle$. Define $d_{t,\min} := \sqrt{\|a_t\|^2 - \alpha_t^2}$.*
3. *Choose a target deviation $d_t \geq d_{t,\min}$ (feasibility).*
4. *Set the CCDS coefficient by the explicit root $\lambda_t := \alpha_t - \sqrt{d_t^2 - d_{t,\min}^2}$, (or the $+$ root, or any deterministic root-selection rule).*
5. *Update $\widehat{y}_t := \widehat{y}_{t-1} + \lambda_t u_t$ and $r_t := y - \widehat{y}_t$.*
6. Dictionary-coordinate update. *Let $c^{(t)}$ and $\beta^{(t)}$ defined as in (14) and (15) respectively.*

The intrinsic CCDS output is $\widehat{y}_T \in Y_T$. Equivalently, the same output admits the explicit selected-atom representation

$$\widehat{y}_T = \sum_{j=1}^{T} \beta_j^{(T)} q_j \in \operatorname{span}\{q_1, \ldots, q_T\} \subset \operatorname{span} \mathcal{D}.$$

Corollary 23 (Finite-dimensional feasibility obstruction for CCDS) *According to Lemma 19, if for some step t the prescribed target deviation satisfies $d_t < d_{t,\min}$, then there exists no scalar $\lambda_t \in \mathbb{R}$ such that $\rho(r_{t-1} - \lambda_t u_t, Y_{t-1}) = d_t$. Consequently, any deviation profile violating the stepwise feasibility thresholds is unrealizable in the fixed ambient realization space.*

The following example shows that in finite dimension, scaffold exhaustion can force u_t to be nearly orthogonal to the residual direction a, producing a strictly positive feasibility threshold $d_{\min}$. Certain deviation targets d_t are then geometrically impossible.

Example 4 (Finite-dimensional infeasibility and necessity of dimension lifting) *Consider a single CCDS step with ambient Hilbert space $H = \mathbb{R}^2$, current scaffold subspace $Y_{t-1} = \{0\}$, one data point $x_1 = e_1 = (1, 0)$, target operator $A = I$, so the residual at step $t-1$ is $r_{t-1}(x_1) = Ax_1 = e_1$, and off–scaffold component $a := (I - \Pi_{Y_{t-1}}) r_{t-1}(x_1) = e_1$, with $\|a\| = 1$.*

Assume that the current scaffold dictionary is exhausted in the sense that the only available innovation direction is $u_t = e_2 = (0,1)$*, so that the update direction is forced to lie in the one–dimensional subspace* $\mathcal{U}_t = \operatorname{span}\{e_2\} \subset Y_{t-1}^{\perp}$*. Hence, Lemma 19 (feasibility bound) gives*

$$d_{\min} = \sqrt{\|a\|^2 - \langle a, u_t\rangle^2} = \sqrt{1-0} = 1,$$

so for all target deviations $d_t < 1$ *the CCDS step is infeasible in the current finite–dimensional realization space: no choice of* λ *can achieve the target deviation because the available innovation direction is orthogonal to the off–scaffold component* a*.*

Adding one new orthogonal dimension restores feasibility by allowing a new innovation direction with controlled alignment to a. This justifies the need for dimension lifting in order to guarantee universal deviation feasibility and, consequently, collapse of the parametrization error in CCDS.

Theorem 24 (One-step feasibility by dimension lifting) *Let* H *be a real Hilbert space,* $Y \subset H$ *a closed subspace, and* $r \in H$*. Define* $a := (I - \Pi_Y)r \in Y^{\perp}$*. Fix a target deviation* d *satisfying* $0 \le d \le \|a\|$*. Then there exists a Hilbert space* $\widetilde{H} := H \oplus \mathbb{R}$*, an isometric embedding* $\iota : H \to \widetilde{H}$*, and a unit vector* $u \in (\iota(Y))^{\perp} \subset \widetilde{H}$ *such that the CCDS distance equation* $\rho(\iota(r) - \lambda u,\, \iota(Y)) = d$ *admits a real solution* $\lambda \in \mathbb{R}$*. Moreover, one can choose* u *so that the minimal achievable deviation is exactly* $d_{\min}(u) = d$*.*

Proof Define $\widetilde{H} := H \oplus \mathbb{R}$ with inner product

$$\langle (x,s),(y,t)\rangle_{\widetilde{H}} := \langle x,y\rangle_H + st,$$

and the isometric embedding $\iota(x) = (x,0)$. Then $\iota(Y)$ is a closed subspace and

$$(\iota(Y))^{\perp} = Y^{\perp} \oplus \mathbb{R}.$$

Let

$$\widetilde{r} := \iota(r), \qquad \widetilde{a} := (I - \Pi_{\iota(Y)})\widetilde{r} = (a,0),$$

so that $\|\widetilde{a}\| = A$. If $A = 0$, then $d = 0$ and the claim is trivial. Assume $A > 0$ and define $e_1 := \frac{\widetilde{a}}{A}$, $e_2 := (0,1)$. Then $e_1, e_2 \in (\iota(Y))^{\perp}$ are orthonormal. Choose $\theta \in [0,\pi/2]$ such that $\sin\theta = \frac{d}{A}$. Define $u := \cos\theta\, e_1 + \sin\theta\, e_2$. Then $u \in (\iota(Y))^{\perp}$ and $\|u\| = 1$. Moreover, $\alpha := \langle \widetilde{a}, u\rangle_{\widetilde{H}} = A\cos\theta$.

By the CCDS feasibility formula (Lemma 3.3),

$$d_{\min}(u) = \sqrt{\|\widetilde{a}\|^2 - \alpha^2} = \sqrt{A^2 - A^2\cos^2\theta} = A\sin\theta = d.$$

Hence $d \ge d_{\min}(u)$, and the equation $\rho(\widetilde{r} - \lambda u, \iota(Y)) = d$ admits a real solution λ. This completes the proof. ■

Remark 25 *To enforce the same target deviation level* d *simultaneously for* M *sample-wise residuals* $\{r^{(j)}\}_{j=1}^{M}$ *at a given CCDS step, it suffices to iterate the one–step dimension–lifting construction of Theorem 24 for each residual. This yields an isometric embedding into an ambient Hilbert space of dimension at most* $\dim(H) + M$ *in which the deviation constraint* $\rho(r^{(j)} - \lambda u, Y) = d$ *is feasible for all* $j = 1, \ldots, M$*.*

Resolution of Example 4 by lifting one dimension. Embed H isometrically into $\widetilde{H} = \mathbb{R}^3$ by $x \mapsto (x, 0)$, and extend the orthogonal complement with the new basis vector e_3. Choose the lifted innovation direction $\tilde{u}_t = \cos\theta\, e_1 + \sin\theta\, e_3, \qquad \sin\theta = d_t$. Then $\tilde{u}_t \in (\iota(Y_{t-1}))^\perp$, $\|\tilde{u}_t\| = 1$, and $\langle \tilde{a}, \tilde{u}_t\rangle = \cos\theta$, $d_{\min}(\tilde{u}_t) = \sqrt{1 - \cos^2\theta} = \sin\theta = d_t$. Hence the CCDS equation

$$\rho\big(\iota(r_{t-1}(x_1)) - \lambda\tilde{u}_t,\, \iota(Y_{t-1})\big) = d_t$$

admits a real solution, and the deviation d_t becomes feasible after adding a single orthogonal dimension.

Remark 26 *At this point, it is natural to clarify why no direct analogue of the CCDS lifting theorem appears in the fixed-V scaffold-selection analysis. In CCDS, lifting is natural because the obstruction arises from having fixed an innovation direction u that is too poorly aligned with the residual geometry; enlarging the ambient space creates new admissible innovation directions and may therefore restore feasibility. In scaffold selection, by contrast, once the admissible scaffold space V is fixed, one already optimizes over all admissible directions in $S = V \cap U^\perp$. Thus there is no further local directional repair inside the same V left to exploit: infeasibility outside the interval $[\lambda_{\min}, \lambda_{\max}]$ reflects a limitation of the fixed admissible scaffold space itself. A possible remedy would be to replace V by a larger admissible scaffold space. This does not lie outside the general framework of the paper, but it belongs to a different level of the theory, namely adaptive modification of the admissible scaffold class rather than deviation control inside a fixed one.*

Greedy residual-alignment as the boundary case of CCDS. At Phase 3, CCDS constructs updates of the form $r_t = r_{t-1} - \lambda_t u_t$, where $u_t \in Y_{t-1}^\perp$ is an innovation direction and λ_t is selected by solving a one–dimensional distance equation $\rho(r_{t-1} - \lambda_t u_t, Y_{t-1}) = d_t$, for a feasible deviation target $d_t \geq d_{t,\min}$.

Consider the trivial scaffold $Y = \{0\}$. Then $\Pi_Y = 0$ and $\rho(r, Y) = \|r\|$. Innovation directions coincide with dictionary atoms: $u = q \in \mathcal{D}$. The CCDS distance equation reduces to $\|r_{t-1} - \lambda_t q_t\| = d_t$, with feasibility interval

$$d_t \in \Big[\min_{\lambda} \|r_{t-1} - \lambda_t q_t\|,\ \|r_{t-1}\|\Big].$$

Lemma 27 is a classical fact from Hilbert space geometry (orthogonal projection onto a one–dimensional subspace). We include a short proof for completeness and to make the paper self-contained.

Lemma 27 (Minimal–deviation update) *For fixed $r \in \mathcal{H}$ and unit $q \in \mathcal{H}$, the function $\lambda \mapsto \|r - \lambda q\|^2$ attains its unique minimum at $\lambda^\star = \langle r, q\rangle$, with minimum value $d_{t,min} = \|r\|^2 - |\langle r, q\rangle|^2$.*

Proof Let $r \in H$ and let $q \in H$ with $\|q\| = 1$. For $\lambda \in \mathbb{R}$ we expand

$$\|r - \lambda q\|^2 = \langle r - \lambda q, r - \lambda q\rangle = \|r\|^2 - 2\lambda\langle r, q\rangle + \lambda^2.$$

This quadratic polynomial in λ attains its minimum at $\lambda^\star = \langle r, q\rangle$. Evaluating at this value gives $\min_{\lambda\in\mathbb{R}} \|r - \lambda q\|^2 = \|r\|^2 - \langle r, q\rangle^2$. Hence the optimal coefficient is $\lambda^\star = \langle r, q\rangle$ and the minimal deviation is $d_{\min} = \sqrt{\|r\|^2 - \langle r, q\rangle^2}$. ■

Thus, choosing the deviation target $d_t = d_{t,\min}$ and selecting the corresponding minimizing root yields the update $r_t = r_{t-1} - \langle r_{t-1}, q_t\rangle q_t$, and the CCDS framework collapses into the Matching Pursuit algorithm. We refer to this Phase–3 realization regime as the *Matching Pursuit regime (MP)*. Thus, MP is defined as the specialization of CCDS to the trivial scaffold $Y = \{0\}$ (no nested subspace scaffolds) with minimal deviation targets $d_t = d_{t,\min}$, in which the general deviation equation reduces to greedy contraction and geometric residual decay.

Remark 28 *MP therefore appears as the boundary case of a deviation–equation solver on nested-subspaces orbit scaffolds.*

So we have.

Proposition 29 *MP is exactly CCDS restricted to the trivial scaffold $Y = \{0\}$ with minimal–deviation target selection.*

Remark 30 ***Why allow*** $d_t > d_{\min}$***?*** *The freedom to choose $d_t > d_{\min}$ is not merely numerical. It is what distinguishes CCDS from the extremal greedy choice $d_t = d_{\min}$, which forces the largest one-step contraction. Allowing $d_t > d_{\min}$ makes it possible to prescribe non-extremal feasible residual profiles, improves numerical stability by avoiding boundary-root behavior in the distance equation, and helps prevent repeated near-collinearity in successive innovation directions. In this sense, the flexibility $d_t > d_{\min}$ is one of the main sources of the robustness of CCDS in weak-alignment or degenerate regimes, as shown in our experiments, see Section 4.*

Interpretability. Our realizability framework is interpretable end-to-end. Phase 1^{real} selects the scaffold, learns an explicit shift operator W, and produces verifiable interpolation maps J_k. Phase 2^{real} then constructs sparse realizations via CCDS, with explicit coefficients and certified deviation control. As a result, the realization mechanism is directly inspectable at the level of scaffold geometry, active atoms, and solver behavior. Full interpretability diagnostics, including supports, atom-usage patterns, coefficient traces, deviation paths, and stopping statistics, are reported in the complementary materials.

Relation between scaffold selection and CCDS realization mechanism. The scaffold selection mechanism developed above should be read as the subspace-level counterpart of the CCDS philosophy introduced later for realizability. In scaffold selection, the controlled object is the projection-error functional, and the admissible moves are nested one-dimensional scaffold expansions inside the orbit-constrained geometry induced by the data. In the realizability stage, the controlled object is instead the parametrization residual, and the admissible moves are coefficient updates along innovation directions. The two mechanisms should therefore be viewed as parallel deviation-control theories attached to the two main bottlenecks of the framework: scaffold geometry and constructive realization. We present scaffold selection first because the logical order of the framework is geometric before constructive: one first selects the support on which the target can be meaningfully projected, then constructs a realizer on that support. In this sense, scaffold selection is the operator-level and more structural manifestation of the same CCDS viewpoint, while the realizability algorithm developed next is its coefficient-level counterpart. The obstruction below (Theorem 31) prescribes the full local residual-covariance operator governing both the remaining projection energy and the best admissible one-step drop. This is specific to the scaffold stage since the deviation viewpoint becomes operator-valued. Thus the result should be read as a scaffold-level strengthening of the classical Bernstein lethargy philosophy, rather than as a phenomenon absent from coefficient-level CCDS.

A Bernstein-type obstruction to uniform scaffold collapse. The constructive results above show that, along a nested admissible scaffold chain, every *stepwise feasible* projection-energy profile can be enforced exactly. A complementary question is whether the scaffold-level deviation-control formalism, by itself, yields any nontrivial uniform rate of collapse for the projection term. The answer is negative. In the spirit of Bernstein lethargy phenomena in approximation theory (5; 17), the next result shows that along a fixed admissible chain, one may realize arbitrarily slow residual decay exactly. Thus the preceding constructive theory introdcued in this section is sharp: it characterizes feasibility, but it does not by itself imply a universal decay law.

In the present scaffold-setting, the natural obstruction object is the compressed residual covariance

$$M_t = \Pi_{S_t} R_t R_t^\top \Pi_{S_t}, \qquad S_t := U_t^\perp,$$

since it governs both the residual energy $E_t = \|R_t\|_F^2$ and the local one-step spectral ceiling $\lambda_{\max}(M_t)$. The theorem below shows that, along a fixed admissible chain, one may prescribe this operator-level obstruction exactly by a diagonal construction adapted to the chain.

Theorem 31 (Operator-valued Bernstein lethargy along an admissible scaffold chain) *Let $U_0 \subset U_1 \subset \cdots \subset U_T \subset \mathbb{R}^C$ be a nested chain of subspaces satisfying $\dim(U_t/U_{t-1}) = 1,\ t = 1, \ldots, T$. Let $a_1, \ldots, a_{T+1} \geq 0$ be arbitrary nonnegative numbers. Assume either $a_{T+1} = 0$ or $U_T \neq \mathbb{R}^C$. Then there exist an orthonormal family $w_1, \ldots, w_T, w_{T+1} \in \mathbb{R}^C$ such that*

$$U_t = \operatorname{span}\{w_1, \ldots, w_t\}, \qquad t = 1, \ldots, T,$$

and a snapshot matrix $Y \in \mathbb{R}^{C \times (T+1)}$ such that, for every $t = 0, \ldots, T$, if

$$R_t := (I - \Pi_{U_t})Y, \qquad S_t := U_t^{\perp}, \qquad M_t := \Pi_{S_t} R_t R_t^{\top} \Pi_{S_t},$$

then $R_t R_t^{\top} = \sum_{j=t+1}^{T+1} a_j\, w_j w_j^{\top}$ and hence $M_t = \sum_{j=t+1}^{T+1} a_j\, w_j w_j^{\top}$. Consequently:

1. *the nonzero spectrum of M_t is exactly $\sigma(M_t) \backslash \{0\} = \{a_{t+1}, \ldots, a_{T+1}\}$ counted with multiplicity;*
2. *the residual energy is $E_t = \|R_t\|_F^2 = \operatorname{tr}(M_t) = \sum_{j=t+1}^{T+1} a_j$;*
3. *the greedy maximal one-step drop is $\lambda_{\max}(M_t) = \max_{j=t+1,\ldots,T+1} a_j$.*

Proof For each $t = 1, \ldots, T$, choose a unit vector $w_t \in U_t \cap U_{t-1}^{\perp}$. Since $\dim(U_t/U_{t-1}) = 1$, the family $\{w_1, \ldots, w_T\}$ is orthonormal and

$$U_t = \operatorname{span}\{w_1, \ldots, w_t\}, \qquad t = 1, \ldots, T.$$

If $a_{T+1} > 0$, choose in addition a unit vector $w_{T+1} \in U_T^{\perp}$. This is possible by the assumption $U_T \neq \mathbb{R}^C$. If $a_{T+1} = 0$, the choice of w_{T+1} is immaterial. Define the snapshot matrix

$$Y := \left[\sqrt{a_1}\, w_1, \sqrt{a_2}\, w_2, \ldots, \sqrt{a_{T+1}}\, w_{T+1}\right].$$

Fix $t \in \{0, \ldots, T\}$. Since $\Pi_{U_t} w_j = w_j$ for $j \leq t$ and $\Pi_{U_t} w_j = 0$ for $j > t$, one has

$$R_t = (I - \Pi_{U_t})Y = \left[0, \ldots, 0, \sqrt{a_{t+1}}\, w_{t+1}, \ldots, \sqrt{a_{T+1}}\, w_{T+1}\right].$$

Because the nonzero columns are mutually orthogonal,

$$R_t R_t^{\top} = \sum_{j=t+1}^{T+1} a_j\, w_j w_j^{\top}.$$

Moreover, for $j > t$ one has $w_j \in U_t^{\perp} = S_t$, hence compression by Π_{S_t} leaves this operator unchanged:

$$M_t = \Pi_{S_t} R_t R_t^{\top} \Pi_{S_t} = \sum_{j=t+1}^{T+1} a_j\, w_j w_j^{\top}.$$

Thus the nonzero spectrum of M_t is precisely the multiset $\{a_{t+1}, \ldots, a_{T+1}\}$. Taking the trace gives

$$E_t = \|R_t\|_F^2 = \operatorname{tr}(R_t R_t^{\top}) = \sum_{j=t+1}^{T+1} a_j,$$

and the largest eigenvalue is $\lambda_{\max}(M_t) = \max_{j=t+1,\ldots,T+1} a_j$. This proves the claim. ■

Corollary 32 (Scalar Bernstein lethargy for residual energies) *Let* $E_0 \geq E_1 \geq \cdots \geq E_T \geq 0$ *be any nonincreasing sequence. Under the same geometric assumption on the chain, there exists a snapshot matrix* Y *such that* $\|(I - \Pi_{U_t})Y\|_F^2 = E_t,\ t = 0, \ldots, T.$

Proof Apply Theorem 31 with

$$a_j := E_{j-1} - E_j, \qquad j = 1, \ldots, T, \qquad a_{T+1} := E_T.$$

These numbers are nonnegative, and $\sum_{j=t+1}^{T+1} a_j = E_t$ by telescoping. ■

Corollary 33 (Monotone spectral sharpness) *If, in addition,* $a_1 \geq a_2 \geq \cdots \geq a_{T+1} \geq 0$, *then for every* $t = 0, \ldots, T$, *we have* $\lambda_{\max}(M_t) = a_{t+1}$. *Hence the greedy maximal drop at stage* t *can be prescribed arbitrarily by a monotone sequence.*

Proof Under the monotonicity assumption, $\max_{j=t+1,\ldots,T+1} a_j = a_{t+1}$. The claim therefore follows directly from Theorem 31. ■

2.3 Situating Our Operator-Theoretic Realizability Framework in Machine Learning.

A conceptual motivation behind the deviation-control theories developed in this work comes from Bernstein lethargy theory in approximation theory (5; 17): along nested approximation families, error decay may be arbitrarily slow, so no universal collapse rate can be expected without additional structure. This perspective is closely aligned with our framework. Rather than organizing either scaffold selection or realization around an a priori decay law, we treat the residual profile itself as an object to be controlled. This principle appears in two parallel forms throughout our paper: at the scaffold level, through feasibility intervals and prescribed projection-error profiles; and at the realization level, through CCDS, where one enforces prescribed deviation constraints step by step instead of relying only on monotone greedy descent.

Against this backdrop, it is natural to situate our deviation-control framework relative to the main traditions that also organize approximation through residuals, namely discrepancy principles in inverse problems and greedy pursuit methods in approximation theory.

Residual thresholds are classical in inverse problems, most notably through Morozov's discrepancy principle and its variants, where a target residual level is used as a stopping or regularization-parameter selection criterion rather than as an equality constraint at each iteration (15; 2). By contrast, in our framework the residual schedule is internalized into the update rule itself. At the scaffold level, one prescribes feasible projection-error drops; at the realization level, CCDS enforces a prescribed deviation schedule by solving an explicit one-dimensional distance equation. In both cases, the residual path is not merely monitored, but treated as a first-class object to be prescribed and constructively realized.

Greedy methods such as Matching Pursuit (MP) (13), Orthogonal Matching Pursuit (OMP) (16; 19), and the broader theory of greedy approximation in Banach and Hilbert spaces (9; 18) also organize approximation through residual reduction. But in these approaches the per-step update is defined through an implicit optimization, projection, or descent principle. Our framework instead recasts the update as the constructive realization of a prescribed residual profile: for scaffold selection, this means realizing feasible projection-error trajectories under admissible one-dimensional scaffold expansions; for CCDS, it means realizing feasible deviation profiles under admissible one-dimensional realization updates. To the best of our knowledge, this perspective does not appear in prior greedy pursuit or approximation algorithms. Moreover, classical greedy MP appears as a boundary case of the realization-level mechanism, just as greedy maximal projection collapse appears as a boundary case of the scaffold-level mechanism.

Consequently, our deviation-control framework constitutes a novel conceptual contribution in three primary ways: (i) it is the first to treat the residual path as a prescribed constraint to be solved for, rather than a result to be monitored; (ii) it replaces implicit optimization with explicit synthesis; and (iii) it provides a unified theory that recovers standard greedy algorithms as a specialized subset of a broader deviation-control class governing scaffold selection and CCDS. In this way, our work grounds both scaffold selection and realization in orbit dynamics, providing a constructive and verifiable framework for operator realizability on data-induced scaffolds in which feasible residual trajectories are built into the theory itself.

3 Prediction as extension of constructive orbit-based realizability

The philosophy before formalization. Before entering the technical development of the predictive part of the paper, we explain the guiding idea. Let $\Xi_{train} = \{(x_i, y_i) : 1 \le i \le n\}$ be the training set. In our framework, the training data does not only specify the outputs to be matched; through the constructive realizability stage, it also induces a preferred orbit-based representation of those outputs. Prediction is then built by extending this constructive structure rather than by learning the target map freely.

A natural first attempt would be to use the entire training set as the interpolation set. One could then seek a partition $\Xi_{train} = D_0 \sqcup \cdots \sqcup D_{K-1}$ such that the exact interpolation theorem applies on each block D_k, yielding a weighted shift W and interpolants J_k with $y_i = J_k W^{(L_i)} x_i$ for every training pair (x_i, y_i). This would provide an exact constructive teacher on the full training set and lead to a predictive rule of the form

$$\widetilde{f}(x) = \sum_{k=0}^{K-1} \sum_{m=0}^{d-1} \alpha_{k,m}(x)\, J_k W^{(m)} x.$$

But this full-training regime is not the one we ultimately want: the orbit dictionary becomes highest-dimensional, the model moves closer to memorization, and in noisy settings exact interpolation of the full training set is undesirable.

This leads to the alternative adopted in this work. Rather than building the scaffold from all of Ξ_{train}, we build it from a carefully selected subset. On that subset we again apply exact interpolation, obtaining a shift W and interpolants J_k, whose orbit atoms generate a dictionary $\mathcal{D}$ and scaffold $U = \operatorname{span} \mathcal{D}$. The subset is chosen so that the resulting scaffold captures as much as possible of the relevant output geometry of the full training set.

Once the scaffold has been selected, we do not attempt to interpolate the full target directly. Instead, we project the target onto U and construct on U a deterministic CCDS realizer. This produces an effective coefficient map on the training set. The final predictive step is then to learn, on the full training set, a coefficient map that reproduces and extends this constructive orbit-based rule from the training points to all of $\mathbb{R}^d$.

This is what makes the predictive step different from ordinary supervised fitting. The labels y_i specify what output must be produced, but the CCDS realizer specifies how that output is constructively produced through orbit atoms on the selected scaffold. The predictor is therefore not trained to learn the target map freely; it is trained to reproduce and generalize a structured constructive rule.

Our predictive philosophy is thus organized around three explicit bottlenecks: the scaffold bottleneck, the realizability bottleneck, and the predictive coefficient bottleneck. The first determines how well the selected scaffold captures the target geometry; the second measures how well the projected target can be constructively realized on that scaffold; and the third governs how well the learned predictor reproduces and extends the resulting coefficient rule beyond the training set. In this way, the realizability theory developed in the first part of the paper becomes the structural backbone of prediction rather than a separate component placed alongside it.

Formalization. One of the most striking features of our theory is that the structural decomposition developed for realizability carries over directly to prediction on the dataset. In the predictive setting the target map is simply the dataset correspondence $f(x_i) = y_i$. Theorem 34 below is an in-sample structural theorem that identifies the right bottlenecks for learning on the training data: (i) the quality of the selected scaffold U, through the projection error $\|f - \Pi_U f\|_\Xi$; (ii) the quality of the CCDS realization on that scaffold through the parametrization error; and (iii) the quality of the shallow learned policy through its approximation on the effective CCDS coefficient map on the dataset.

Let $\Xi_{train} = \{(x_i, y_i)\}_{i=1}^n \subset \mathbb{R}^d \times \mathbb{R}^C$, $\Xi_{test} = \{(\bar{x}_j, \bar{y}_j)\}_{j=1}^m \subset \mathbb{R}^d \times \mathbb{R}^C$. Define the dataset seminorm $\|g\|_\Xi := (\sum_{x:(x,y)\in\Xi} \|g(x)\|_{\ell^2})/|\Xi|$ for maps g defined on $\{x : (x, y) \in \Xi\}$.

Theorem 34 (Geometric in-sample prediction bound) *Fix a scaffold $U = span\, \mathcal{D} \subset \mathbb{R}^C$ with set of orbit atoms $\mathcal{D} = \{J_k W^{(m)} x_i : (x_i, y_i) \in \Xi_{train},\ x_i \in D_k,\ 0 \le k \le K-1,\ 0 \le m \le d-1\}$. Define the target map on Ξ_{train} by $f(x_i) := y_i$. And let $f_{\mathrm{CCDS}}(x) := \sum_{k=0}^{K-1} \sum_{m=0}^{d-1} a^{\mathrm{eff}}_{k,m}(x)\, J_k W^{(m)}(x)$, with $x \in U$, be the CCDS realizer on the scaffold U. Denote $a^{\mathrm{eff}}(x) := \big(a^{\mathrm{eff}}_{k,m}(x)\big)_{k,m} \in \mathbb{R}^{Kd}$. Assume our predictor has the form*

$$\widetilde{f}(x) := \sum_{k=0}^{K-1} \sum_{m=0}^{d-1} \alpha_{k,m}(x)\, J_k W^{(m)}(x) \tag{16}$$

and satisfies, for each $i = 1, \ldots, n$, $\|\alpha(x_i) - a^{\mathrm{eff}}(x_i)\|_{\ell^1} \le \delta_i$. Then

$$\|f - \widetilde{f}\|_{\Xi_{train}} \le \underbrace{\|f - \Pi_U f\|_{\Xi_{train}}}_{\varepsilon_{proj}(f,U,\Xi_{train})} + \underbrace{\|\Pi_U f - f_{\mathrm{CCDS}}\|_{\Xi_{train}}}_{\varepsilon_{param}(f,U,\Xi_{train})} + M\,\bar{\delta},$$

where $\bar{\delta} := \frac{1}{n}\sum_{i=1}^n \delta_i$ and $M := \max_{0\le k\le K-1,\ 0\le m\le d-1,\ 1\le i\le n} \|J_k W^{(m)}(x_i)\|_{\ell^2} < \infty$ by boundedness of the operators $J_k W^{(m)}$.

Proof Fix $i \in \{1, \ldots, n\}$. Then

$$(\Pi_U f)(x_i) - \widetilde{f}(x_i) = \big((\Pi_U f)(x_i) - f_{\mathrm{CCDS}}(x_i)\big) + \big(f_{\mathrm{CCDS}}(x_i) - \widetilde{f}(x_i)\big).$$

Hence, by the triangle inequality,

$$\|(\Pi_U f)(x_i) - \widetilde{f}(x_i)\|_{\ell^2} \le \|(\Pi_U f)(x_i) - f_{\mathrm{CCDS}}(x_i)\|_{\ell^2} + \|f_{\mathrm{CCDS}}(x_i) - \widetilde{f}(x_i)\|_{\ell^2}.$$

We estimate the second term. By definition,

$$f_{\mathrm{CCDS}}(x_i) - \widetilde{f}(x_i) = \sum_{k=0}^{K-1} \sum_{m=0}^{d-1} \big(a^{\mathrm{eff}}_{k,m}(x_i) - \alpha_{k,m}(x_i)\big)\, J_k W^{(m)} x_i.$$

Applying the triangle inequality again,

$$\|f_{\mathrm{CCDS}}(x_i) - \widetilde{f}(x_i)\|_{\ell^2} \le \sum_{k=0}^{K-1} \sum_{m=0}^{d-1} \big|a^{\mathrm{eff}}_{k,m}(x_i) - \alpha_{k,m}(x_i)\big|\, \|J_k W^{(m)} x_i\|_{\ell^2}.$$

Then,

$$\|f_{\mathrm{CCDS}}(x_i) - \widetilde{f}(x_i)\|_{\ell^2} \le M \sum_{k=0}^{K-1} \sum_{m=0}^{d-1} \big|a^{\mathrm{eff}}_{k,m}(x_i) - \alpha_{k,m}(x_i)\big|.$$

That is, $\|f_{\mathrm{CCDS}}(x_i) - \widetilde{f}(x_i)\|_{\ell^2} \le M\, \|\alpha(x_i) - a^{\mathrm{eff}}(x_i)\|_{\ell^1} \le M\, \delta_i$. Therefore,

$$\|(\Pi_U f)(x_i) - \widetilde{f}(x_i)\|_{\ell^2} \le \|(\Pi_U f)(x_i) - f_{\mathrm{CCDS}}(x_i)\|_{\ell^2} + M\, \delta_i.$$

Averaging over $i = 1, \dots, n$, we obtain

$$\|\Pi_U f - \widetilde{f}\|_{\Xi_{train}} \le \|\Pi_U f - f_{\text{CCDS}}\|_{\Xi_{train}} + M\,\bar{\delta}.$$

Finally, observe that $f - \widetilde{f} = (f - \Pi_U f) + (\Pi_U f - \widetilde{f})$. Applying the triangle inequality,

$$\|f - \widetilde{f}\|_{\Xi_{train}} \le \|f - \Pi_U f\|_{\Xi_{train}} + \|\Pi_U f - \widetilde{f}\|_{\Xi_{train}}.$$

Substituting the bound just proved yields

$$\|f - \widetilde{f}\|_{\Xi_{train}} \le \|f - \Pi_U f\|_{\Xi_{train}} + \|\Pi_U f - f_{\text{CCDS}}\|_{\Xi_{train}} + M\,\bar{\delta}.$$

This completes the proof. ∎

Theorem 34 shows that the in-sample predictive instantiation of our orbit-based framework, which we call Weighted Backward Shift Neural Networks (WBSNNs) and defined formally below in Definition 38, is naturally governed by three bottlenecks that define the three phases of the model. The first is the construction of an optimal scaffold, which determines the projection quality and therefore defines Phase 1. The second bottleneck, and hence Phase 2, is the CCDS realizability of the projected targets on that scaffold. Phase 3 is the shallow learned policy that approximates the effective CCDS coefficient map on the dataset, thereby controlling the final parametrization gap. In this sense, prediction in WBSNN inherits the same structural logic as realizability, but with an additional learned coefficient-approximation layer in the final phase.

3.1 Predictive Framework (WBSNNs).

We now formalize WBSNN Phases $1^{\text{pred}} - 3^{\text{pred}}$.

Phase 1^{pred}: Budgeted predictive scaffold selection in WBSNN. Let $\Xi_{train} = \{(x_i, y_i)\}_{i=1}^n$ be the training set. For $b \in \mathcal{B} \subset (0,1]$, set $m_b := \lfloor bn \rfloor$ and $\mathfrak{X}_b := \{\Xi_{\text{sub}} \subset \Xi_{train} : \ |\Xi_{\text{sub}}| \le m_b\}$.

Fix $\Xi_{\text{sub}} \in \mathfrak{X}_b$, write $M := |\Xi_{\text{sub}}|$, and let $c \in \mathcal{C} \subseteq \{1, \dots, M\}$. A *cap-constrained admissible partition* of Ξ_{sub} is a decomposition $\Xi_{\text{sub}} = D_0 \sqcup \cdots \sqcup D_{K-1}$ such that $1 \le |D_k| \le c$ for all k, the recursive linear independence condition (4) holds inside each D_k, and $K-1$ is minimal among such partitions; denote this minimum by $K(\Xi_{\text{sub}}, c) - 1$. The blocks need not be homogeneous.

Given (Ξ_{sub}, c), apply exact interpolation Theorem 4 or via regularized/pseudoinverse on each D_k. This yields a learned weighted shift $W_{\Xi_{\text{sub}},c}$ and local interpolants $J_{0,\Xi_{\text{sub}},c}, \ \dots, \ J_{K(\Xi_{\text{sub}}-1,c),\Xi_{\text{sub}},c}$. Define the orbit dictionary

$$\mathcal{D}(\Xi_{\text{sub}}, c) := \{J_{k,\Xi_{\text{sub}},c} W^{(m)}_{\Xi_{\text{sub}},c} x_i : \ (x_i, y_i) \in D_k, \ 0 \le k \le K(\Xi_{\text{sub}}, c) - 1, \ 0 \le m \le d-1\},$$

and the induced scaffold $U(\Xi_{\text{sub}}, c) := \operatorname{span} \mathcal{D}(\Xi_{\text{sub}}, c) \subset \mathbb{R}^C$.

For each budget b, the primary geometric criterion is the certified projection error obtained by applying the scaffold-selection algorithm from subsection 2.1 among all the scaffolds varying $\Xi_{\text{sub}} \in \mathfrak{X}_b$, and $c \in \mathcal{C}$ and get one selected scaffold $U_b^\star := U(\Xi^\star_{\text{sub}}(b), c^\star(b))$. Let $f(x_i) = y_i$, for all $(x_i, y_i) \in \Xi_{train}$, be the target function, and define

$$\varepsilon_{proj}(b) := \frac{\sum_{(x_i,y_i)\in\Xi^*_{\text{sub}}(b)} ||f(x_i) - \Pi_{U_b^*} f(x_i)||_2}{|\Xi^*_{\text{sub}}(b)|}, \qquad \text{where } \Pi_U f(x_i) = UU^T f(x_i).$$

Let $\varepsilon_{min} := \min_{b\in\mathcal{B}} \varepsilon_{proj}(b)$ and given $\delta > 0$ form the budget δ-neighborhood $\mathcal{B}_\delta := \{b : \varepsilon_{proj}(b) \le \varepsilon_{min} + \delta\}$.

This first stage is justified since in previous sections we identified scaffold geometry as a genuine bottleneck.

Empirical Observation (implicit regularization). Under vanishing regime, the following decay estimate underlies an intrinsic denoising behaviour and controls the intrinsic memory horizon. Proof of Lemma 35 deferred to the Appendix **??**.

Lemma 35 (Noise Suppression via Orbit Decay) *Let $z \in \mathbb{R}^d$ and let $W^{(m)}$ with weights $w_0, \dots, w_{d-1} \in \mathbb{R}$ such that $\rho := \prod_{i=0}^{d-1} |w_i|$. Then for all $m \geq 1$, $\|W^{(m)} z\| \leq \rho^{\lfloor m/d \rfloor} \cdot C$, where $C := \max_{0 \leq r \leq d-1} \|W^{(r)} z\|$. In particular, if $\rho < 1$ and z is an additive noise vector, then $\|W^{(m)} z\| \to 0$ exponentially as $m \to \infty$.*

This yields noise suppression without explicit penalties (like dropout layers), driven solely by the orbit geometry of the shift operator itself. The result is stable learning behavior in noisy or low-sample settings. Together, these effects give WBSNN a transparent mechanism for orbit stability, effective rank, and noise attenuation under the vanishing regime. More interpretable than standard neural architectures, whose contraction behavior has no comparable closed-form description.

Thus, Lemma 35 fully justifies that our next priority in selecting the final scaffold is vanishing regimes. If the weights of $W_{\Xi^*_{\text{sub}}(b), c^*(b)}$ are $w_0, \dots, w_{d-1}$, set $\rho(b) := \prod_{i=0}^{d-1} |w_i|$. Now define the vanishing-budget subset $\mathcal{B}^{van}_\delta := \{b \in \mathcal{B}_\delta : \rho(b) < 1\}$.

Case 1: If $|\mathcal{B}^{van}_\delta| = 1$, then b^* is the unique element of $\mathcal{B}_\delta$. The final scaffold is $U^*_{b^*}$. This rule gives strict priority to vanishing dynamics whenever projection quality is already essentially tied, which is consistent with the role of vanishing orbits as an implicit regularizer in WBSNN as proved in Lemma 35.

Case 2: If $|\mathcal{B}^{van}_\delta| > 1$, then among the budgets in $\mathcal{B}^{van}_\delta$ we select the scaffold minimizing a secondary score based on conditioning of the interpolants J_k, effective rank and compatibility with the vanishing regime as justified by Lemma 35.

So, the final selection rule should be $U^*_{b^*}$, where $b^* \in \arg\min_{b \in \mathcal{B}^{van}_\epsilon} \text{Score}_{van}(b)$, and

$$\text{Score}_{van}(b) := \omega_{\text{van}} \rho(b) + \omega_{\text{cond}} \kappa^{norm}(b) + \omega_{\text{eff}} r^{norm}_{\text{eff}}(b; \varepsilon),$$

with nonnegative weights summing to 1 . Here:

(i) Interpolation conditioning. If $\mathcal{M}_k(\Xi^*_{\text{sub}}(b), c^*(b))$ is the local linear system used to build $J_{k, \Xi^*_{\text{sub}}(b), c^*(b)}$ on D_k (exactly or via a regularized/pseudoinverse solve in practice), define the interpolation conditioning of block D_k by $\kappa(\mathcal{M}_k) := \|\mathcal{M}_k\|_2 \|\mathcal{M}_k^\dagger\|_2$, if $\mathcal{M}_k$ has full column rank, otherwise set $\kappa(\mathcal{M}_k) = \infty$, where $\mathcal{M}_k^\dagger$ is the Moore-Penrose pseudoinverse of $\mathcal{M}_k$. So $\kappa(\mathcal{M}_k)$ measures how stable the local interpolation problem is on D_k, where smaller $\kappa(\mathcal{M}_k)$ means the local interpolation system is better conditioned, hence more numerically stable. The budget-level conditioning score is then

$$\kappa(b) := \max_{1 \leq k \leq K(\Xi_{\text{sub}}, c)} \kappa(\mathcal{M}_k(\Xi^*_{\text{sub}}(b), c^*(b))).$$

We then consider the normalized version: define $\kappa_{min} := \min_{b \in \mathcal{B}^{van}_\delta}$, $\kappa_{max} := \max_{b \in \mathcal{B}^{van}_\delta}$. If $\kappa_{max} > \kappa_{min}$, then set $\kappa^{norm}(b) := (\kappa(b) - \kappa_{min})(\kappa_{max} - \kappa_{min})$, otherwise set $\kappa^{norm}(b) := 0$.

(ii) Effective rank. Fix a tolerance $\varepsilon > 0$. For $(x, y) \in D_k$, define the effective horizon relative to x by

$$M_{\text{eff}}(x; \varepsilon) := \min\Big\{M \in \mathbb{N} : \big\|J_{k, \Xi^*_{\text{sub}}(b), c^*(b)} W^{(m)}_{\Xi^*_{\text{sub}}(b), c^*(b)} x\big\| \leq \varepsilon \text{ for all } m \geq M\Big\}.$$

Assume $\rho(b) < 1$, then by the decay estimate of Lemma 35 we have the explicit bound

$$M_{\text{eff}}(x; \varepsilon) \; < \; d \left\lceil \frac{\log(\varepsilon / C)}{\log \; \rho(b)} \right\rceil \tag{17}$$

where $C := \max_{0 \leq r \leq d-1} \|J_{k, \Xi^*_{\text{sub}}(b), c^*(b)} W^{(r)}_{\Xi^*_{\text{sub}}(b), c^*(b)} x\|$. In the vanishing regime, this measures the number of geometrically active orbit directions before deeper iterates become negligible. Proof of (17) deferred to the Appendix **??**. In particular, the *the pointwise effective rank* defined by

$$r_{\text{eff}}(x; b, \varepsilon) := \dim \text{span}\{J_{k, \Xi^*_{\text{sub}}(b), c^*(b)} W^{(m)}_{\Xi^*_{\text{sub}}(b), c^*(b)} x : \; (x, y) \in D_k, \; 0 \leq m < M_{\text{eff}}(x; \varepsilon)\},$$

is structurally bounded by the orbit dynamics. Finally, we define *the subset-level effective rank* by

$$r_{\text{eff}}(b;\varepsilon) := \frac{1}{|\Xi_{\text{sub}}|} \sum_{(x_i,y_i)\in\Xi^*_{\text{sub}}(b)} r_{\text{eff}}(x_i;b,\varepsilon),$$

In other words, $r_{\text{eff}}(b;\varepsilon)$ is the average number of geometrically active orbit directions before vanishing dynamics make deeper iterates negligible. So it measures intrinsic orbit complexity. We then consider the normalized version: set $r_{min} := \min_{b\in\mathcal{B}^{van}_{\delta}} r_{\text{eff}}(b;\varepsilon)$ and $r_{max} := \max_{b\in\mathcal{B}^{van}_{\delta}} r_{\text{eff}}(b;\varepsilon)$. If $r_{max} > r_{min}$, then set $r^{norm}_{\text{eff}}(b;\varepsilon) := (r_{\text{eff}}(b;\varepsilon) - r_{min})/(r_{max} - r_{min})$, otherwise set $r^{norm}_{\text{eff}}(b;\varepsilon) := 0$.

Thus, in the vanishing regime, smaller effective horizon means fewer active orbit directions, which is exactly the structural simplicity we want to reward together with interpolation conditioning once projection error is already essentially tied. We preferred normalization because it makes the weights interpretable since each term is in $[0,1]$.

Case 3: If $|\mathcal{B}^{van}_{\delta}| = \emptyset$, then among the budgets in $\mathcal{B}_{\delta}$ we select the scaffold minimizing a penalized score $\text{Score}_{novan}(b)$. Thus, the final selection rule should be $U^*_{b^*}$, where $b^* \in \arg\min_{b\in\mathcal{B}_{\epsilon}} \text{Score}_{novan}(b)$, and

$$\text{Score}_{novan}(b) := \omega_{\text{novan}} V^{norm}_{novan}(b) + \omega_{\text{cond}}\kappa^{norm}(b),$$

where $V_{novan}(b) := \max\{0, \rho(b) - 1\}$, $V_{min} := \min_{b\in\mathcal{B}_{\delta}} V_{novan}(b)$ and $V_{max} := \max_{b\in\mathcal{B}_{\delta}} V_{novan}(b)$. If $V_{max} > V_{min}$, then define $V^{norm}_{novan}(b) := (V_{novan}(b) - V_{min})/(V_{max} - V_{min})$, otherwise set $V^{norm}_{novan}(b) := 0$. Thus, the non-vanishing penalty should increase as $\rho(b)$ moves away from the vanishing regime.

Phase 2^{pred}: CCDS realizability on the Phase 1^{pred} selected scaffold. Let

$$f_{\text{CCDS}}(x) := \sum_{k=0}^{K-1}\sum_{m=0}^{d-1} a^{\text{eff}}_{k,m}(x)\, J_k W^{(m)}(x), \qquad (x,y)\in\Xi_{train}$$

be the CCDS realizer on the optimal scaffold $U^*_{b^*}$ from Phase 1^{pred}. Denote $a^{\text{eff}}(x) := \big(a^{\text{eff}}_{k,m}(x)\big)_{k,m} \in \mathbb{R}^{Kd}$.

Phase 3^{pred}: Shallow learned policy approximating f_{CCDS}. Mimicking dynamics in infinite dimensions, we define Phase 3^{pred} of WBSNN as a generalization mechanism over the orbit-induced class

$$\widetilde{f}(x) = \sum_{k=0}^{K-1}\sum_{m=0}^{\infty} \alpha_{k,m}(x)\, J_k W^{(m)}(x), \qquad x\in\mathbb{R}^d,$$

trained on the full dataset Ξ_{train}. Equivalently, Phase 3^{pred} learns over the family generated by the orbit geometry associated with all training samples, while producing a predictor defined for arbitrary inputs $x \in \mathbb{R}^d$. Note that this is the reason why we considered unwrapping $W^{(L)}$ modulo d; otherwise, standard backward shifts would vanish beyond depth d, preventing generalization along the infinite orbit. By Lemma 3, in finite-dimensional spaces our generalization expression collapses—up to scalar factors to the prediction rule:

$$\widetilde{f}(x) = \sum_{k=0}^{K-1}\sum_{m=0}^{d-1} \alpha_{k,m}(x)\, J_k W^{(m)} x, \qquad x\in\mathbb{R}^d. \tag{18}$$

In order to be able to apply the in-sample transfer Theorem 34, the Phase 3^{pred} predictor must be explicitly of the form prescribed by the prediction rule (18), and the associated coefficient map must approximate the effective CCDS coefficient map on that scaffold. Thus, Phase 3^{pred} is not trained to learn coefficients from scratch, but to approximate f_{CCDS} within the same orbit-based representation class.

The next step is to define the predictor $\widetilde{f}$ explicitly so that: (i) it has the form prescribed by the prediction rule (18); (ii) it is endowed with an explicit guiding structure that allows it to track the CCDS coefficient map effectively; and (iii) it is implemented through a hybrid shallow policy, thereby preserving interpretability.

Definition 36 (Predictive Hybrid Learned CCDS mechanism of Phase 3^{pred}) *Assuming the outputs of Phases 1^{pred} and 2^{pred} have already been obtained, we now introduce the hybrid Phase 3^{pred} mechanism, which is trained on the full dataset Ξ_{train} and yields a predictor defined on all of $\mathbb{R}^d$.*

Orbit dictionary: *For each input $x \in \mathbb{R}^d$, define the orbit atoms*

$$d_{k,m}(x) := J_k W^{(m)} x \in \mathbb{R}^C, \qquad k \in \{0, \dots, K-1\}, \;\; m \in \{0, \dots, d-1\},$$

and their normalized versions

$$\widetilde{d}_{k,m}(x) := \frac{d_{k,m}(x)}{\|d_{k,m}(x)\| + \varepsilon_{\text{norm}}},$$

where $\varepsilon_{\text{norm}} > 0$ is fixed.

We write $\mathcal{D}(x) := \{\widetilde{d}_{k,m}(x) : 0 \le k \le K-1,\; 0 \le m \le d-1\}$.

Learned policy variables: *Let $T \ge 1$ be the number of Phase 3^{pred} steps. For each $t = 1, \dots, T$, let $g_\theta^{(t)} : \mathbb{R}^{d+C} \to \mathbb{R}^{Kd}$, $h_\theta^{(t)} : \mathbb{R}^{d+C} \to \mathbb{R}$, be shallow networks (e.g. one-hidden-layer MLPs). Their role is only to select the* policy variables *of the CCDS step: a soft atom-mixture and a feasible deviation slack. The CCDS coefficient itself remains explicit.*

We also let $b_\theta : \mathbb{R}^d \to \mathbb{R}^C$ be a shallow initialization map, used to define the residual target for the constructive recursion.

Initialization: *Set $\widehat{y}_0(x) := 0$, $r_0(x) := b_\theta(x)$, $Y_0(x) := \{0\}$.*

Step $t = 1, \dots, T$. Assume $\widehat{y}_{t-1}(x)$, $r_{t-1}(x)$, and $Y_{t-1}(x)$ are already defined. Define the state

$$z_t(x) := \big[x,\; r_{t-1}(x)\big] \in \mathbb{R}^{d+C}.$$

Soft policy over orbit atoms: *Let $\pi_t(x) := \operatorname{softmax}\big(g_\theta^{(t)}(z_t(x))\big) \in \Delta^{Kd-1}$, and define the soft selected direction $q_t(x) := \sum_{k=0}^{K-1} \sum_{m=0}^{d-1} \pi_{t;k,m}(x)\, \widetilde{d}_{k,m}(x)$.*

Innovation direction: *Let $v_t(x) := q_t(x) - \Pi_{Y_{t-1}(x)} q_t(x)$. Assuming $\|v_t(x)\| \neq 0$, define*

$$u_t(x) := \frac{v_t(x)}{\|v_t(x)\| + \varepsilon_{\text{orth}}}, \qquad Y_t(x) := Y_{t-1}(x) \oplus \operatorname{span}\{u_t(x)\},$$

where $\varepsilon_{\text{orth}} > 0$ is fixed.

Feasible deviation target: *Set $a_t(x) := (I - \Pi_{Y_{t-1}(x)})\, r_{t-1}(x)$, $\beta_t(x) := \langle a_t(x), u_t(x) \rangle$, and $d_{t,\min}(x) := \sqrt{\max\big(\|a_t(x)\|^2 - \beta_t(x)^2,\, 0\big)}$. Let $\Delta_t(x) := \operatorname{softplus}\big(h_\theta^{(t)}(z_t(x))\big) \ge 0$, $d_t(x) := d_{t,\min}(x) + \Delta_t(x)$. Hence $d_t(x) \ge d_{t,\min}(x)$ by construction, so feasibility is automatic.*

Explicit CCDS coefficient. *Define the step coefficient by the CCDS root*

$$\lambda_t(x) := \beta_t(x) - \sqrt{\max\big(d_t(x)^2 - d_{t,\min}(x)^2,\, 0\big)}.$$

(One may also choose the $+$ root if desired; throughout we fix one branch consistently.)

Residual update: *Set $\widehat{y}_t(x) := \widehat{y}_{t-1}(x) + \lambda_t(x)\, u_t(x)$, $r_t(x) := b_\theta(x) - \widehat{y}_t(x)$.*

Final predictor: *The Phase 3 output is $\widetilde{f}(x) := \widehat{y}_T(x)$.*

By Proposition 21, $\widetilde{f}$ is of the form prescribed by (18), then we can apply the in-sample transfer Theorem 34 to our predictor $\widetilde{f}$ from Definition 36. So Phase 3^{pred} should be understood as a mechanism for constructing the effective coefficients in a structured and interpretable way, namely as an approximation of the CCDS coefficient map associated with the scaffold and atoms dictionary produced by Phases 1^{pred} and 2^{pred}.

Coefficient-supervised Phase–3^{pred} training. To enforce both prediction accuracy and coefficient tracking, we train Phase 3^{pred} by combining the usual prediction loss with a coefficient-supervision loss against the effective CCDS coefficients. The resulting Phase 3^{pred} loss is

$$\mathcal{L}_{\text{Phase 3}} = \mathcal{L}_{\text{pred}} + \lambda_{\text{coeff}} \frac{1}{n} \sum_{i=1}^{n} \|\alpha(x_i) - \alpha^{\text{eff}}(x_i)\|_{\ell^2}.$$

Thus, the in-sample predictive bound of Theorem 34 becomes exactly:

$$\|f - \widetilde{f}\|_{\Xi_{train}} \leq \underbrace{\|f - \Pi_{U^*_{b^*}} f\|_{\Xi_{train}}}_{\varepsilon_{proj}(f, U^*_{b^*}, \Xi_{train})} + \underbrace{\|\Pi_{U^*_{b^*}} f - f_{\text{CCDS}}\|_{\Xi_{train}}}_{\varepsilon_{param}(f_{\text{CCDS}}, U^*_{b^*}, \Xi_{train})} + M \, \frac{1}{n} \sum_{i=1}^{n} \|a^{\text{eff}}(x_i) - \alpha(x_i)\|_{\ell^1}, \quad (19)$$

where $f(x_i) = y_i$ for all $(x_i, y_i) \in \Xi_{train}$; $M = \max_{0 \leq k \leq K-1,\ 0 \leq m \leq d-1, 1 \leq i \leq n} \|J_k W^{(m)}(x_i)\|_{\ell^2}$; $a^{\text{eff}}(x_i) = (a^{\text{eff}}_{k,m}(x_i))_{k,m} \in \mathbb{R}^{Kd}$ and $\alpha(x_i) = (\alpha_{k,m}(x_i))_{k,m} \in \mathbb{R}^{Kd}$.

Hence, in-sample prediction succeeds when the scaffold selected in Phase 1^{pred} is good enough that the projected target is close to the true dataset map, the Phase 2^{pred} CCDS realizer on that scaffold is good enough to provide the model with a structured surrogate of the projected map, so that the target map is not learned in an unconstrained way, and finally the hybrid Phase 3^{pred} policy is good enough to track the CCDS coefficient map.

Remark 37 *The reader should note that:*

(i) *Noise Suppression via Orbit Decay Lemma 35 shows that although Phase* 3^{pred} *in WBSNN only uses orbit powers* $m \leq d-1$*, these terms implicitly encode decay across the full infinite orbit* $\{W^{(m)}\varepsilon\}_{m \geq 0}$ *via scalar rescaling thanks to Lemma 3. The vanishing regime* $\prod_{i=0}^{d-1} w_i < 1$ *induces exponential suppression of long-term perturbations, acting as an intrinsic regularizer in the orbit geometry.*

(ii) *Phase* 3^{pred} *is deliberately defined on infinite orbits in order to pave the way for the extension of WBSNNs to infinite-dimensional spaces. This is fully natural in our framework, since both the exact interpolation Theorem 4 and the CCDS mechanism hold in infinite-dimensional Banach spaces. In this paper, however, we restrict our attention to the finite-dimensional case; a full treatment of the infinite-dimensional WBSNN framework is left for future work.*

We are now ready to formally introduce WBSNNs in the finite-dimensional setting.

Definition 38 (WBSNN on a finite-dimensional dataset) *Given a finite-dimensional training dataset* Ξ_{train}*, a* Weighted Backward Shift Neural Network (WBSNN) *associated to* Ξ_{train} *is the predictor* $\widetilde{f}$ *obtained from the three-phase pipeline of this paper: Phase* 1^{pred} *selects a scaffold* $U^*_{b^*}$*, Phase* 2^{pred} *constructs on that scaffold a CCDS realizer of the projected target, and Phase* 3^{pred} *learns a data-dependent coefficient map* $\alpha_{k,m}(x)$ *through the hybrid predictive CCDS mechanism from Definition 36, yielding the final prediction rule* $\widetilde{f}$ *via* (18).

Moreover, the predictor $\widetilde{f}$ from Definition 38 satisfies on Ξ_{train} the in-sample prediction error bound given in (19).

Remark 39 *WBSNN is a general-purpose learning paradigm that operates across modalities and tasks without requiring task-specific architectural redesign. It replaces stacked nonlinearities with structured orbit dynamics. Moreover, Theorem 34 is only an in-sample structural result: it does not by itself establish test generalization. That question is addressed later in Theorem 40. Nevertheless,* *the framework already suggests a structural inductive bias favorable to robust generalization: optimal scaffold budgets limit capacity, the orbit-span constraint restricts the hypothesis class, vanishing dynamics suppress noise, relaxed interpolation improves stability in noisy regimes, and the shallow Phase* 3^{pred} *head only reweights orbit atoms rather than learning an unrestricted predictor.*

Under the additional assumption that test points lie near the same orbit geometry captured at training, we obtain an out-of-sample prediction bound that preserves the same structural logic as the in-sample bound from Theorem 34. Thus, although the WBSNN architecture already exhibits a structural inductive bias favorable to generalization, an explicit out-of-sample result is still needed to complete the predictive picture of the framework.

Let $U \subset \mathbb{R}^C$ be the scaffold selected from Ξ_{train}, and let $f : \mathbb{R}^d \to \mathbb{R}^C$ be the target map, so that $f(x_i) = y_i,\ f(\bar{x}_j) = \bar{y}_j$. Let $\widetilde{f}$ be the predictor built on the scaffold U.

Theorem 40 (Geometric out-of-sample prediction bound) *Assume*

1. *(Neighborhood relation between test and train) There exists* $\delta \geq 0$ *such that for every* $\bar{x}_j \in \Xi_{test}$ *there exists some* $x_{i(j)} \in \Xi_{train}$ *with* $\|\bar{x}_j - x_{i(j)}\|_2 \leq \delta$.

2. *(Regularity of the predictor) The map* $\widetilde{f}$ *is* $L_{\widetilde{f}}$*-Lipschitz on the same neighborhood:* $\|\widetilde{f}(x) - \widetilde{f}(x')\|_2 \leq L_{\widetilde{f}}\|x - x'\|^2_{\ell},\ \forall x,\ x' \in \Xi_{train} \cup \Xi_{test}$.

Then

$$\|f - \widetilde{f}\|_{\Xi_{test}} \leq \underbrace{\|f - \Pi_U f\|_{\Xi_{test}}}_{\varepsilon_{proj}(f,U,\Xi_{test})} + \max_{x:(x,y)\in\Xi_{train}} \|\Pi_U fx - \widetilde{f}x\|_{\ell^2} + (1 + L_{\widetilde{f}})\delta.$$

Proof Fix $j \in \{1, \ldots, m\}$, and choose $x_{i(j)} \in \Xi_{train}$ such that $\|\bar{x}_j - x_{i(j)}\|_{\ell^2} \leq \delta$. Then

$$f(\bar{x}_j) - \widetilde{f}(\bar{x}_j) = \big(f(\bar{x}_j) - Pf(\bar{x}_j)\big) + \big(\Pi_U f(\bar{x}_j) - \Pi_U f(x_{i(j)})\big) + \big(\Pi_U f(x_{i(j)}) - \widetilde{f}(x_{i(j)})\big) + \big(\widetilde{f}(x_{i(j)}) - \widetilde{f}(\bar{x}_j)\big).$$

Applying the triangle inequality,

$$\begin{aligned}\|f(\bar{x}_j) - \widetilde{f}(\bar{x}_j)\|_2 \leq{}& \|f(\bar{x}_j) - \Pi_U f(\bar{x}_j)\|_2 + \|\Pi_U f(\bar{x}_j) - \Pi_U f(x_{i(j)})\|_2 \\ &+ \|\Pi_U f(x_{i(j)}) - \widetilde{f}(x_{i(j)})\|_2 + \|\widetilde{f}(x_{i(j)}) - \widetilde{f}(\bar{x}_j)\|_2.\end{aligned}$$

We have, $\|\Pi_U f(\bar{x}_j) - \Pi_U f(x_{i(j)})\|_2 \leq \|\Pi_U\|\delta = \delta$. By Assumption 2, $\|\widetilde{f}(x_{i(j)}) - \widetilde{f}(\bar{x}_j)\|_{\ell^2} \leq L_{\widetilde{f}}\delta$. Therefore,

$$\|f(\bar{x}_j) - \widetilde{f}(\bar{x}_j)\|_{\ell^2} \leq \|f(\bar{x}_j) - \Pi_U f(\bar{x}_j)\|_{\ell^2} + \|\Pi_U f(x_{i(j)}) - \widetilde{f}(x_{i(j)})\|_{\ell^2} + (1 + L_{\widetilde{f}})\delta.$$

Averaging over $j = 1, \ldots, m$ we obtain

$$\|f - \widetilde{f}\|_{\Xi_{test}} \leq \|f - \Pi_U f\|_{\Xi_{test}} + \max_{x:(x,y)\in\Xi_{train}} \|\Pi_U fx - \widetilde{f}x\|_{\ell^2} + (1 + L_{\widetilde{f}})\delta.$$

This proves the theorem. ■

Remark 41 *The out-of-sample version preserves the same logic as the in-sample version: (i)* $\|f - \Pi_U f\|_{\Xi_{test}}$ *is the scaffold bottleneck on test; (ii)* $\max_{x:(x,y)\in\Xi_{train}} \|\Pi_U f(x) - \widetilde{f}(x)\|_{\ell^2}$ *is the predictor bottleneck inherited from training; and (iii)* $(1 + L_{\widetilde{f}})\delta$ *is the new transfer term, which says that test points must lie near the training scaffold geometry. If* $\delta = 0$*, Theorem 40 collapses back to the same two bottlenecks: projection error on test and parametrization error of the predictor on training.*

Remark 42 (On the Lipschitzness of Phase 3^{pred} predictor) *The Lipschitz assumption on $\widetilde{f}$ in Theorem 40 is essentially already built into Definition 36. Indeed, on any bounded region $\Omega \subset X$, the orbit atoms $x \mapsto J_k W^{(m)} x$ are bounded linear maps, hence Lipschitz. The learned objects in Phase–3^{pred} are the primitive neural maps appearing in Definition 36; all other quantities z_t, π_t, q_t, v_t, u_t, β_t, $d_{t,\min}$, Δ_t, d_t, λ_t, $\widehat{y}_t$, r_t are induced recursively from these learned maps, the orbit atoms, and fixed algebraic operations. Moreover, the softened normalization $u_t(x) = \frac{v_t(x)}{\|v_t(x)\|+\varepsilon_{\text{orth}}}$ removes the only singularity that would otherwise arise from orthogonalization at $v_t(x) = 0$. Thus, in the present formulation, the only remaining possible obstruction to Lipschitz continuity comes from the CCDS root term $\lambda_t(x) = \beta_t(x) - \sqrt{\max\big(d_t(x)^2 - d_{t,\min}(x)^2, 0\big)}$. Accordingly, it suffices to impose the uniform feasibility margin*

$$d_t(x)^2 - d_{t,\min}(x)^2 \geq \eta_0^2 > 0 \qquad \textit{for all } x \in \Omega \textit{ and all } t.$$

This holds, for instance, if the learned slack $\Delta_t(x) = d_t(x) - d_{t,\min}(x)$ is uniformly bounded away from zero on Ω and $d_{t,\min}$ is uniformly bounded there. Under this condition, the square-root term is Lipschitz on Ω, hence so is λ_t. By finite composition of Lipschitz maps, the residual recursion and therefore the final hybrid predictor $\widetilde{f}$ are Lipschitz on Ω. Thus the additional hypothesis needed for the out-of-sample theorem is not a structural change to Phase–3^{pred}, but only a uniform positive feasibility margin.

On $\alpha_{k,m}$ modalities. Concerning the MLP-learned coefficient $\alpha_{k,m}$ of our prediction rule (18), we consider four WBSNN modalities along two axes—head expressivity and backbone sharing. **(1)** $\alpha_{k,m}$: a full per-k, per-state head that gates all dynamical states m within each subset k (maximal flexibility). **(2)** $\alpha_{k,L}$: a middle-ground head that gates only the states established by Phase 1^{pred} and 2^{pred}, meaning for each $0 \leq k \leq K-1$, the coefficient $\alpha_{k,L} = 0$ for all $L \notin \mathcal{L}_k$, where $\mathcal{L}_k$ is the set of L_i satisfying condition (4) for $X_i \in D_k$, reducing variance while retaining within-k selectivity. **(3)** α_k (shared backbone): a per-k head trained on a trunk *primed by* $\alpha_{k,m}$; operationally it aggregates over m using features learned for the richer head, offering strong generalization with lower overfitting. **(4)** α_k (separate backbones): the same per-k head trained on its own trunk, decoupled from $\alpha_{k,m}$. So, $\alpha_{k,L}$ is an intermediate head that sits between $\alpha_{k,m}$ and α_k. Our main protocol uses variants (1) and (3), i.e. full $\alpha_{k,m}$ and α_k with a shared, $\alpha_{k,m}$-primed backbone, balancing accuracy, robustness, and efficiency. Experiments report the behavior of $\alpha_{k,m}$ and both α_k variants across datasets. We reserve the variant $\alpha_{k,L}$ exclusively for the infinite-dimensional formulation of WBSNNs (future work), where it plays a central role.

Interpolation modalities and practical considerations. While Theorem 4 gives a constructive proof of exact interpolation via sequential updates of the operator J, our implementation follows the same underlying principle while allowing both exact and relaxed interpolation, depending on the numerical regime. In noiseless or well-conditioned settings, we use exact interpolation; in noisy, compressed, or ill-conditioned regimes, we allow a relaxed variant for stability and robustness. This distinction is important conceptually: the exact modality reflects the theorem in its cleanest form, whereas the relaxed modality shows how the same orbit-based mechanism behaves under more realistic conditions. In practice, for learning W, we search over depths $L \leq d$ and test span membership through a least-squares residual with tolerance τ, which serves as a numerically efficient proxy for the recursive independence condition (4) in Theorem 4. To keep the procedure tractable, we optimize W on mini-subsets and form the blocks D_k from a subsampled portion of the training set. The interpolation operators J_k are computed in one of three ways: by the constructive rank-one updates from the proof of Theorem 4, by the closed-form pseudoinverse when the relevant orbit matrix is full-rank, or by a regularized pseudoinverse in noisy or ill-conditioned cases. Our main experiments default to the regularized form in such regimes, while exact variants are used when the independence structure is sufficiently clean. This gives a unified view of interpolation in the framework: exact

Table 1: Interpretability diagnostics for Phases 1^{pred} and 2^{pred} of WBSNN. Lower projection residual and lower interpolation/parametrization errors indicate a scaffold that captures the target geometry well and supports accurate constructive realization. Higher explained label-space energy indicates that a larger fraction of the target lives in the selected scaffold. Values are mean ± standard deviation over seeds.

Dataset	$\varepsilon_{\text{proj}} \downarrow$	$E_{\text{expl}} \uparrow$	Anchor residual ↓	$\varepsilon_{\text{param}} \downarrow$
MNIST	0.19 ± 0.03	0.84 ± 0.03	0.012 ± 0.004	0.021 ± 0.006
IMDb	0.23 ± 0.04	0.79 ± 0.04	0.015 ± 0.005	0.028 ± 0.008
Gas Sensor Drift	0.24 ± 0.04	0.77 ± 0.05	0.019 ± 0.006	0.031 ± 0.009

Table 2: Interpretability diagnostics for Phase 3^{pred} of WBSNN. Lower coefficient-tracking error and smaller effective support indicate closer adherence to the constructive CCDS rule. Higher top-3 mass and higher neighbor support overlap indicate stronger coefficient concentration and greater local support stability. Values are mean ± standard deviation over seeds.

Dataset	$E_{\text{coeff}} \downarrow$	Support size ↓	Top-3 mass ↑	Neighbor overlap ↑
MNIST	0.084 ± 0.012	4.1 ± 0.6	0.81 ± 0.05	0.73 ± 0.04
IMDb	0.097 ± 0.015	5.3 ± 0.8	0.76 ± 0.04	0.68 ± 0.05
Gas Sensor Drift	0.091 ± 0.013	4.7 ± 0.7	0.79 ± 0.04	0.71 ± 0.04

interpolation is the canonical theoretical modality, and relaxed interpolation is its stable practical counterpart.

3.2 Interpretability of the full WBSNN pipeline

A central claim of the framework is that WBSNN is interpretable end-to-end. This interpretability is phasewise and structural. In Phase 1^{pred}, the model selects an explicit data-induced orbit scaffold, so the geometric support on which learning takes place is visible rather than latent. In Phase 2^{pred}, the projected target is constructively realized on that scaffold through blockwise interpolation and CCDS-based synthesis, so the realizability stage is governed by explicit certificates rather than by opaque hidden representations. In Phase 3^{pred}, prediction is not learned as unrestricted feature mixing, but as a data-dependent coefficient law over the orbit atoms generated by the scaffold and the interpolation operators. Thus each phase carries its own interpretable object: scaffold geometry, constructive realization, and coefficient law.

To make this end-to-end interpretability measurable, we report diagnostics aligned with the three phases of WBSNN. For Phase 1^{pred}, we use the projection residual

$$\varepsilon_{\text{proj}} = \frac{1}{\sqrt{n}} \|(I - \Pi_U)Y\|_F$$

and the explained label-space energy

$$E_{\text{expl}} = \frac{\|\Pi_U Y\|_F^2}{\|Y\|_F^2},$$

which quantify how much of the target geometry is captured by the selected scaffold. For Phase 2^{pred}, we report the anchor interpolation residual and the post-CCDS parametrization error, which measure how accurately the projected target is realized once the scaffold has been fixed. For Phase 3^{pred}, we report the coefficient-tracking error against the effective CCDS coefficients, the effective support size of the learned coefficient vector, the top-3 coefficient mass, and the support-overlap stability across nearby test points. Together, these diagnostics make it possible to inspect the full WBSNN pipeline rather than only its final outputs.

More precisely, let $\Xi_{\text{anc}} = \{(x_i, y_i)\}_{i=1}^{n_{\text{anc}}}$ denote the anchor/discovery set selected in Phases 1^{pred}–2^{pred}, let $U := U(\Xi_{\text{anc}})$ be the selected scaffold, and let $\widehat{f}^{(2)}$ denote the Phase 2^{pred} realizer on U.

We define the *anchor interpolation residual* by

$$R_{\mathrm{anc}} := \frac{1}{n_{\mathrm{anc}}} \sum_{i=1}^{n_{\mathrm{anc}}} \big\| y_i - \widehat{f}^{(2)}(x_i) \big\|_2,$$

which measures how accurately the Phase 2^{pred} construction fits the anchor labels themselves. Since Phase 2^{pred} is meant to realize the *projected* target on the selected scaffold, we also define the *post-CCDS parametrization error*

$$\varepsilon^{\mathrm{pred}}_{\mathrm{param}} := \frac{1}{n_{\mathrm{anc}}} \sum_{i=1}^{n_{\mathrm{anc}}} \big\| \widetilde{y}_i - \widehat{f}^{(2)}(x_i) \big\|_2, \qquad \widetilde{y}_i := \Pi_U y_i,$$

which isolates the constructive realization error once the scaffold has already been fixed.

For Phase 3^{pred}, let $\alpha(x) \in \mathbb{R}^{Kd}$ denote the learned coefficient vector and $\alpha^{\mathrm{eff}}(x) \in \mathbb{R}^{Kd}$ the effective CCDS coefficient vector used as the constructive target in the hybrid loss. We define the *coefficient-tracking error* by

$$E_{\mathrm{coeff}} := \frac{1}{n} \sum_{i=1}^{n} \big\| \alpha(x_i) - \alpha^{\mathrm{eff}}(x_i) \big\|_2,$$

which measures how closely the learned predictive phase tracks the underlying CCDS rule.

To quantify sparsity, fix a threshold $\tau_{\mathrm{supp}} > 0$ and define the *effective support size* by

$$S_{\mathrm{eff}} := \frac{1}{n} \sum_{i=1}^{n} \#\big\{ j : \; |\alpha_j(x_i)| > \tau_{\mathrm{supp}} \big\}.$$

Thus S_{eff} measures how many orbit atoms are meaningfully active on average in the final prediction.

To quantify coefficient concentration, let $\mathrm{Top}_3(\alpha(x_i))$ denote the set of indices of the three largest coefficients in magnitude. We define the *top-3 coefficient mass* by

$$M_3 := \frac{1}{n} \sum_{i=1}^{n} \frac{\sum_{j \in \mathrm{Top}_3(\alpha(x_i))} |\alpha_j(x_i)|}{\sum_{j=1}^{Kd} |\alpha_j(x_i)|},$$

so that larger values indicate that most of the predictive mass is carried by only a few orbit atoms.

Finally, to measure local structural stability, fix an integer $s \geq 1$ and let

$$\mathcal{S}_s(x_i) := \mathrm{Top}_s(\alpha(x_i))$$

be the top-s support of the coefficient vector. If x_i^{nn} denotes a nearest neighbor of x_i in the test set (or in the relevant feature space), we define the *neighbor support-overlap stability* by

$$O_{\mathrm{nn}} := \frac{1}{n} \sum_{i=1}^{n} \frac{\big| \mathcal{S}_s(x_i) \cap \mathcal{S}_s(x_i^{\mathrm{nn}}) \big|}{\big| \mathcal{S}_s(x_i) \cup \mathcal{S}_s(x_i^{\mathrm{nn}}) \big|}.$$

High values of O_{nn} indicate that nearby inputs tend to activate similar scaffold-orbit supports, which is consistent with a locally stable and geometrically meaningful coefficient field.

Table 1 summarizes the geometric and realizability diagnostics of Phases 1^{pred} and 2^{pred}, while Table 2 summarizes the predictive coefficient diagnostics of Phase 3^{pred}. Across all representative datasets, the same qualitative pattern is observed: the selected scaffold captures a large portion of the target geometry, the Phase 2^{pred} realizer fits that projected target with low residual, and the final hybrid predictive phase remains sparse, concentrated, and close to the effective CCDS rule. This supports the intended interpretation of WBSNN as a structured orbit-based pipeline rather than as a black-box predictor.

These two tables make the end-to-end interpretability claim of WBSNN concrete: Phase 1^{pred} exposes the geometry on which learning is performed, Phase 2^{pred} exposes how the projected target is constructively synthesized on that geometry, and Phase 3^{pred} exposes the coefficient law through which prediction is ultimately produced.

3.3 Situating WBSNNs in Machine Learning.

Operator-theoretic ideas have entered machine learning through several distinct routes, including Koopman-inspired approaches that analyze nonlinear dynamical systems through linear operator representations and spectral decompositions (14; 12; 6), operator-learning architectures such as DeepONet that are designed to learn nonlinear operators from data (11), methods such as neural integral equations that learn integral operators from data (21), and architectural modifications aimed at efficiency or stability, such as Linformer, which approximates self-attention by a low-rank mechanism to achieve linear complexity, and NAIS-Net, which derives stable deep architectures from non-autonomous differential equations (20; 7). WBSNNs are conceptually different from all of these. They are not built to approximate nonlinear operators in the usual neural-operator sense, nor to impose low-rank or spectral constraints on an otherwise black-box architecture. Instead, they build prediction from orbit iterates of a single orbit-generating linear operator and organize learning through three linked ingredients: scaffold selection, constructive realizability on the selected scaffold, and extension of the resulting coefficient law. Their mathematical motivation comes from linear dynamics, where iterates of a single operator generate both rich geometric and dynamical behavior, with the weighted backward shift serving as the guiding prototype (3).

4 Experiments

We evaluate the framework in two stages. First, we study *realizability*, where the goal is to test whether a target operator can be constructively synthesized on a selected orbit scaffold, and to isolate the roles of projection quality and deviation-controlled realization. Second, we study *predictability*, where the same scaffold-realization mechanism is used inside WBSNN, with Phase 3^{pred} implemented by the hybrid CCDS coefficient-tracking policy rather than by an unconstrained MLP head. Throughout, the experiments are designed to test the theory's structural claims, not merely to maximize benchmark scores.

Unless otherwise stated, all features are standardized using training statistics only. All main predictive experiments use PCA-compressed features before Phase 1^{pred}, precisely because this regime makes recovery of a useful orbit scaffold a strictly harder task allowing us cleanly test the scaffold and realizability bottlenecks predicted by the theory; raw high-dimensional results follow the same pipeline confirming competitive performance without compression and are discussed in the complementary materials. Only a small *discovery fraction* of the training set is used to build the orbit scaffold. The predictive head uses the hybrid CCDS Phase 3^{pred} mechanism, so that prediction is interpreted as an approximation of the effective CCDS coefficient map rather than as unrestricted function fitting. All experiments run in a lightweight CPU-only Jupyter environment—no GPUs—so the numbers reflect true data/geometry leverage rather than horsepower. The experiments are therefore not designed around a large-scale engineering scaling narrative, but around illustrating the practical consequences of the theory in controlled finite-dimensional regimes through scaffold quality, constructive realizability, and predictive coefficient tracking.

4.1 Experiments on realizability

The realizability experiments isolate the two bottlenecks from Section 2.1: the *projection bottleneck*, governed by scaffold quality, and the *parametrization bottleneck*, governed by constructive synthesis on the selected scaffold. The first two experiments already expose the necessity of CCDS under weak or degenerate geometry. We then add one diagnostic experiment whose sole purpose is to make scaffold quality visible as an empirical object.

Phase 3^{real} protocol. For realizability experiments, Phase 3^{real} uses the full CCDS solver on the selected scaffold. In well-aligned cases, the fast MP/CCDS variant is also reported for comparison, but the full nested-subspace CCDS solver is treated as the reference implementation whenever greedy contraction becomes unstable.

Experiment 1: Pathological weak-alignment stress test and deviation-control ablation. We evaluate Phase 2^{real} realizability on a synthetic dataset constructed to destroy global alignment: each sample is generated in an independent random orthogonal frame and re-rotated, with strong label noise, yielding residuals in generic position relative to any fixed orbit dictionary. This places the solver in a small-cosine, thin-feasible-cone regime where monotone MP contraction is not guaranteed.

A Phase 1^{real} scaffold is frozen under a strict budget ($d = 10$, $M = 700$, random 10% subset, operator family ill-conditioned). Interpolation is deliberately relaxed to accommodate noise, so Phase 1^{real} leaves a dominant projection error ($\varepsilon_{\mathrm{proj}} \approx 0.64$), and Phase 2^{real} must perform genuine innovation.

An ablation over the deviation-control parameter $\gamma_{\mathrm{dev}} \in \{0, 0.5, 0.9\}$ shows that the constructive realizability rate remains stable at $\approx$ 36–37%, indicating that the expressivity ceiling is determined by scaffold geometry rather than by the deviation schedule. In contrast, solver stability depends strongly on γ_{dev}: small values enforce $d_t \approx d_{\min}$ and yield near-optimal parametrization error ($\varepsilon_{\mathrm{param}} \sim 10^{-7}$) with the fast MP/CCDS solver, while large γ_{dev} produces thin feasible caps, stalls greedy alignment, and requires fallback to full CCDS. The latter recovers the same realizability rate, confirming that deviation control governs numerical stability and contraction, not the intrinsic realizable set. Residuals of order 10^{-2} under full CCDS reflect the imposed deviation schedule and finite step budget, not the minimal achievable error within the fixed scaffold. Despite the adversarial geometry, the solutions remain highly interpretable: representations are extremely sparse (median support size $|S_i| = 2$, global sparsity $> 94\%$) and supported on a small, stable subset of orbit atoms with consistent class-conditional usage. Most non-realizable samples terminate by reaching the step limit rather than by infeasibility, showing that the dominant limitation is scaffold incompleteness and projection geometry, not solver instability. We refer the reader to Table 7 and the complementary materials.

Experiment 2: Extreme-Ray Dominance vs. Interior Feasibility in Near-Parallel Subspaces under Weak Scaffolds. We evaluate Phase 2^{real} realizability of MP/CCDS on a challenging synthetic dataset formed by a union of six nearly parallel 3-dimensional subspaces in $\mathbb{R}^{35}$ ($\sigma_{\mathrm{parallel}} = 0.02$, $\sigma_{\mathrm{noise}} = 0.05$), reduced to $d = 15$ by PCA (98.2% variance retained). Labels are generated by linear operators C followed by a fixed readout B, yielding classification tasks with identical input geometry but diverse operator spectra (identity, orthogonal, ill-conditioned, low-rank, spiked, sparse, Toeplitz, and adversarially scaffold-misaligned operator). A deliberately weak, unoptimized random scaffold (3% budget) is used to decouple Phase 2^{real} realizability from scaffold selection.

Despite poor projection coverage, fast MP/CCDS consistently achieves extremely small parametrization error ($\varepsilon_{\mathrm{param}} \sim 10^{-8}$–$10^{-6}$) via very sparse representations. However, strict MP contraction is almost never certified: residuals typically lie on flat or symmetric faces of the feasible cone, where several atoms have nearly equal correlations and no single extreme ray has a margin. Full CCDS, enforcing deviation-controlled multi-atom feasibility rather than one-step contraction, achieves moderate to high realizability (4–88%) across spectra, constructing stable interior two-atom certificates. More details on full diagnostics and interpretability metrics in Tables 3 and 8

Furthermore, the fast MP/CCDS and full CCDS traces for Sample 0 (Identity and Orthogonal operator), illustrate the sharp contrast between greedy dynamics and deviation-controlled certificates (full details in complementary materials). In fast mode, the Identity operator case converges in 14 steps with occasional CCDS fallbacks, while the Orthogonal operator case requires 50 steps and exhibits strong alternation between a few atoms (e.g., repeated switching between 327 and 194), a clear instance of boundary crawling along a flat cone face. This directly visualizes the margin-free pathology: no single atom dominates, so pure MP progresses slowly and fragily. In contrast, full CCDS always produces a clean two-step interior-feasible certificate. Together, these traces

Table 3: Experiment 2: Near-parallel and symmetric degeneracy benchmark. Full CCDS remains robust when greedy single-ray contraction becomes unstable or vacuous.

Operator	MP (%)	Full CCDS (%)	Fast $\varepsilon_{\text{param}}$ (mean)	Full CCDS $\varepsilon_{\text{param}}$ (mean)	Steps (fast mean)	$\varepsilon_{\text{proj}}$ (mean)
I	0.20	1.20	2.06e-06	3.43e-06	20.76	2.56e-06
ORTHO	0.05	2.85	5.05e-07	2.90e-06	20.97	2.60e-06
ILL_SVD	0.00	54.40	1.31e-07	1.02e-06	21.25	6.89e-07
LOWRANK	0.00	78.75	1.84e-07	7.24e-07	18.11	9.34e-07
SPIKED	0.00	25.00	2.18e-07	1.60e-06	20.43	1.44e-06
SPARSE	0.00	26.70	1.88e-07	1.38e-06	20.64	1.10e-06
TOEPLITZ	0.00	87.70	5.73e-08	6.39e-07	20.76	2.58e-07
ADV_SCAFF	0.05	4.80	6.73e-07	2.56e-06	20.93	2.40e-06

provide concrete evidence that greedy contraction is insufficient in near-parallel geometries, whereas deviation control yields short, stable, and geometrically robust realizations.

Overall, Experiment 2 exposes a core limitation of classical greedy contraction: MP requires strict single-ray dominance and fails almost universally due to margin collapse under symmetry, degeneracy, or near-parallelism. Realizability is instead governed by cone interior width and operator-induced anisotropy. This benchmark cleanly separates approximation quality from contraction geometry and shows that deviation-controlled solvers such as CCDS are essential — not optional — for achieving reliable, stable realizability under weak, unoptimized scaffolds. For a full report of interpretability diagnostics (per-sample supports, atom usage histograms, deviation paths, and stopping statistics), we refer the reader to the complementary materials.

Experiment 3: scaffold-quality diagnostic. This new experiment is designed to empirically validate the scaffold bottleneck itself. We reuse the same synthetic generator as in Experiment 1 so that the operator family, ambient dimension, and noise model remain fixed; only the scaffold construction mechanism is varied. For each of 10 runs, we construct three candidate scaffolds with the same atom budget and the same Phase 2^{real} realization budget: a selected orbit scaffold produced by the scaffold-selection algorithm from Section 2.1, a random orbit scaffold produced from the same orbit-generated family but with a random admissible subset/partition choice, and a random non-orbit based Gaussian scaffold built from normalized Gaussian atoms with the same total number of atoms as the orbit dictionary. For each scaffold U, we compute

$$\varepsilon_{\text{proj}}(U) = \frac{1}{\sqrt{N}}\|(I - \Pi_U)Y\|_F, \qquad E_{\text{expl}}(U) = \frac{\|\Pi_U Y\|_F^2}{\|Y\|_F^2}, \qquad \varepsilon_{\text{param}}(U) = \|\Pi_U A - \hat{y}\|_{\Xi}.$$

The first is the empirical projection residual, the second is the *explained label-space energy*, and the third is the post-CCDS parametrization residual.

Table 9 reports the results. The selected orbit scaffold dominates both comparison scaffolds—the random orbit scaffold and the random non-orbit based Gaussian scaffold—not only in projection quality but also in downstream realizability. This experiment is especially important because it shows directly that orbit-based scaffolds do matter for learning: the gain is not coming merely from dictionary size or ambient dimension, but from the orbit structure itself and from selecting that structure in a data-adaptive way. The systematic gap between the selected orbit scaffold, the random orbit scaffold, and the random non-orbit based Gaussian baseline therefore gives concrete empirical evidence that learnability in our framework is driven by scaffold geometry.

4.2 Experiments on predictability

We now evaluate the predictive instantiation of the framework. All main predictive experiments use the protocol: exact or relaxed interpolation when computing the interpolants J_k in Phase 1^{pred}, depending on the regime, and hybrid CCDS coefficient-tracking with α_k (the shared-trunk head) or $\alpha_{k,m}$ (the richer head gating per subset and per orbit state) in Phase 3^{pred}.

The main paper keeps the predictive section intentionally small: three representative datasets and three focused ablations. This is sufficient to illustrate the theory while leaving broader sweeps to complementary materials. The choice of datasets reflects three distinct roles: chronological distribution shift (Gas Sensor Drift), compressed text with an exact-vs-relaxed interpolation tradeoff (IMDb), and compressed vision with a clear discovery-budget effect (MNIST).

Baselines. We compare against strong baselines organized by family and matched to the regime under consideration, but we deliberately keep the main-paper suite compact. Because all predictive experiments are conducted in PCA-compressed finite-dimensional settings, we restrict attention to controls that are meaningful under the same compressed-feature protocol: one strong linear baseline, one strong kernel baseline, one strong tree ensemble, one shallow neural baseline, and, when the dataset warrants it, one domain-matched baseline tailored to the underlying shift or drift structure. This preserves the family-organized comparison principle without turning the main text into an exhaustive benchmark inventory; broader baseline sweeps are deferred to the complementary materials.

Main predictive datasets and protocol. The main predictive results are reported in Table 4.

Gas Sensor Drift. We use the batch-chronological protocol: training on earlier batches and testing strictly out of time on later batches. Features are standardized on the training split and compressed to $d = 10$ principal components. We use approximately $n \approx 3000$ training samples and a 3% discovery fraction for scaffold selection. This dataset serves as the distribution-shift stress test. Under the selected-scaffold + realization on the scaffold + hybrid-CCDS protocol, WBSNN remains competitive against all baselines and clearly outperforms the shallow baselines under strict chronology.

IMDb. We use mean-pooled GloVe-100 embeddings, then compress to $d = 20$ principal components. The training budget is approximately $n \approx 4000$, with a 5% discovery fraction. This dataset serves as the compressed-text setting, and it is also the cleanest place to test exact versus relaxed interpolation. Despite the erasure of word order and the severe compression, the selected orbit scaffold still supports competitive performance, suggesting that the predictive bottleneck is not merely head capacity but the quality of the scaffold-realization pipeline.

MNIST. We use PCA-compressed pixel features with $d = 15$, approximately $n \approx 5000$ training samples, and a 3% discovery fraction. This serves as the compressed-vision setting and is the most natural place to display the discovery-budget effect. WBSNN is fully competitive without any image-specific architecture, using the same orbit-based pipeline as in all other modalities.

For the full set of interpretability diagnostics introduced in subsection 3.2, we refer the reader to the complementary materials.

Table 4: Main predictive results under the updated protocol: scaffold selection in Phase 1^{pred}, CCDS-based realization in Phase 2^{pred}, and hybrid CCDS coefficient tracking in Phase 3^{pred}. For each dataset we report strong baselines organized by family and matched to the compressed-feature regime: one representative linear, kernel, tree, and shallow neural baseline, plus a domain-matched baseline when relevant. Results are mean ± standard deviation over seeds. Best result in bold.

Dataset	WBSNN	Linear	RBF-SVM	XGBoost	MLP	Domain-matched
MNIST	$\mathbf{95.7 \pm 0.3}$	Lin SVM 94.2 ± 0.3	95.4 ± 0.2	93.6 ± 0.4	94.9 ± 0.3	—
IMDb	$\mathbf{80.8 \pm 0.9}$	LogReg 77.4 ± 0.7	79.6 ± 0.8	78.6 ± 0.8	79.1 ± 0.9	Nyström+SVM 79.8 ± 0.8
Gas Sensor Drift	$\mathbf{69.4 \pm 1.8}$	LogReg 61.2 ± 1.1	66.1 ± 1.4	67.3 ± 1.2	64.8 ± 1.6	CORAL 67.8 ± 1.3

Predictive ablations

Ablation 1: selected scaffold vs. random orbit scaffold vs. random Gaussian scaffold. This ablation is one of the most important in the paper, because it isolates the contribution of scaffold geometry itself. We test whether predictive performance is driven merely by dictionary size and Phase 3^{pred} capacity, or whether it depends in an essential way on having the *right* orbit-based scaffold. We therefore compare, under identical budgets and with the same Phase 3^{pred} head, three cases: (i) the selected orbit scaffold returned by the scaffold-selection algorithm, (ii) a random orbit scaffold drawn from the same orbit-generated family, and (iii) a random Gaussian scaffold with the same atom count but no orbit structure.

Table 5 shows a clear and systematic pattern on both MNIST and Gas Sensor Drift. On MNIST, the selected orbit scaffold reaches 95.7% accuracy, compared with 93.8% for the random orbit scaffold and 91.5% for the Gaussian scaffold, while simultaneously achieving the lowest projection residual (0.19 versus 0.28 and 0.37). On Gas Sensor Drift, the same ordering appears: the selected orbit scaffold attains 69.4% accuracy, compared with 66.2% and 62.9%, again with the smallest projection residual (0.24 versus 0.33 and 0.41). Thus the gain is not explained by budget alone. Even within the orbit-generated family, replacing scaffold selection by a random admissible choice already causes a clear drop, and removing orbit structure altogether degrades performance further. In the Gaussian control, Phases 2^{pred} and 3^{pred} are kept identical at the algorithmic level: CCDS constructs a realization over the Gaussian dictionary, and the predictive head is trained to track the corresponding constructive coefficient map. Thus the ablation isolates the effect of scaffold geometry rather than conflating it with differences in the realization or prediction mechanism. The structural conclusion is therefore unambiguous:

$$\text{selected orbit} \; > \; \text{random orbit} \; > \; \text{Gaussian}.$$

This ordering shows that both ingredients matter. First, *orbit structure* matters: random orbit scaffolds consistently outperform unstructured Gaussian scaffolds, indicating that the orbit-generated family carries genuine predictive content. Second, *scaffold selection* matters: the selected orbit scaffold consistently outperforms a random orbit scaffold of the same size, showing that learnability is not driven by orbit structure in the abstract, but by recovering the orbit geometry that is actually compatible with the dataset. In this sense, the ablation gives direct empirical support to the central claim of the paper: prediction improves when the scaffold captures more of the target action, and this improvement comes from structured, data-adaptive orbit geometry rather than from brute-force expansion of the dictionary.

Ablation 2: exact vs. relaxed interpolation. This ablation tests the bridge from the clean exact interpolation regime of Theorem 4 to the stable practical regime used in harder data conditions. We choose IMDb and Gas Sensor Drift because together they isolate the tradeoff sharply. IMDb is the comparatively cleaner compressed-text setting, where exact interpolation remains competitive and stays close to the structural ideal of the theory. Gas Sensor Drift is the more challenging regime, with chronology, drift, and higher effective noise, where relaxed interpolation is distinctly more stable and performs better. Table 6 therefore highlights the key point: exact interpolation is the canonical modality of the theory, but relaxed interpolation is the practically preferable one once noise and geometric instability become significant. Both are realizations of the same orbit-based interpolation mechanism, differing only in how strictly the interpolation stage is enforced.

Ablation 3: discovery-budget curve. This ablation tests the central claim that the scaffold can be discovered from a very small subset of the training data. We vary the discovery fraction while keeping all other settings fixed, and report both test accuracy and explained label-space energy in Table 10. The resulting pattern is decisive. Increasing the budget from 1% to 3% and then to 5% yields substantial gains, but beyond that point the curve flattens sharply: by 15%, and even more clearly by 25%, both scaffold quality and predictive performance have essentially saturated. This is the key point of the ablation. The gains are strongly front-loaded, indicating that the relevant orbit geometry is recovered early, after which enlarging the discovery budget yields only marginal returns. In particular, the 15% and 25% results make clear that the framework is not benefiting from ever-larger scaffolds, but from identifying the right scaffold structure with relatively little data.

Additional experiments. The purpose of the main text is not to provide maximal experimental breadth, but to isolate the minimal set of experiments needed to validate the scaffold bottleneck, the realizability bottleneck, and the predictive coefficient bottleneck under our theory. To complement the controlled experiments presented in the main text and to further substantiate the general-purpose scalability of the orbit-based framework, we performed additional studies deferred to the complementary materials. These include evaluations on raw high-dimensional inputs without PCA preprocessing, scaling experiments on larger datasets, tests with frozen embeddings from large pre-trained models, computational scaling analyses (including weight-regime ablations), a preliminary operator-learning task in infinite-dimensional sequence spaces, and cross-domain scaffold transfer experiments. These results confirm that the WBSNN pipeline and orbit-scaffold mechanism remain effective, interpretable, and structurally consistent when moving beyond the compressed regimes of the main text, reinforcing the framework's general-purpose applicability.

Table 5: Scaffold ablation. We compare the selected orbit scaffold against a random orbit scaffold with the same budget and against a random Gaussian scaffold with the same atom count. The selected orbit scaffold yields both lower projection residual and stronger predictive performance.

Dataset / Scaffold type	Test accuracy (%) ↑	Projection residual ↓	Relative note
MNIST / Selected orbit	**95.7**	**0.19**	Best structured scaffold
MNIST / Random orbit	93.8	0.28	Orbit family without selection
MNIST / Gaussian	91.5	0.37	Unstructured baseline
Gas Sensor Drift / Selected orbit	**69.4**	**0.24**	Best structured scaffold
Gas Sensor Drift / Random orbit	66.2	0.33	Orbit family without selection
Gas Sensor Drift / Gaussian	62.9	0.41	Unstructured baseline

Table 6: Exact versus relaxed interpolation in Phase 2_{pred}. Exact interpolation is the clean structural regime, while relaxed interpolation is often the more stable practical regime under noise, compression, or drift.

Dataset / Phase 2 mode	Test accuracy (%) ↑	Anchor residual ↓	Seed std. ↓	Regime summary
IMDb / Exact	80.4	**0.000**	1.1	Clean structural regime
IMDb / Relaxed	**80.8**	0.011	**0.9**	More stable in practice
Gas Sensor Drift / Exact	66.8	**0.000**	2.4	Sensitive under drift
Gas Sensor Drift / Relaxed	**69.4**	0.019	**1.8**	Stable under drift/noise

Conclusion. The orbit-scaffold perspective suggests a broader research program: datasets should be probed for their intrinsic shift geometry before any parametric model is trained, turning learnability into a question of geometric compatibility rather than unrestricted capacity.

5 Future Work.

The present framework suggests several natural next directions.

Extension to infinite dimensions. While this paper develops the theory primarily in finite dimensions, its core constructive mechanisms already persist in infinite-dimensional Banach settings. A natural continuation is to develop the full infinite-dimensional predictive theory and study orbit-based learning on genuinely functional data, including settings in which the native objects are signals, fields, or operators rather than finite vectors.

Toward a geometric theory of datasets. More broadly, this work suggests the existence of dataset-specific configurations $(K_{\mathcal{D}}, W_{\mathcal{D}}, J_{\mathcal{D}})$ such that WBSNN acts as a functor extracting the intrinsic geometry of $\mathcal{D}$. This perspective may ultimately allow us to reason about learning not only through data and models, but through the geometry they induce. If formalized, it could ground a geometric or category-theoretic framework for learning, in which orbit representations encode generalization structure.

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Appendix A. Remaining Proofs and Tables

Proof of Lemma 35. By Lemma 3, for every $m \geq 1$, there exists a scalar $\lambda_m > 0$ such that

$$W^{(m)}z = \lambda_m \cdot W^{(m \bmod d)}z.$$

Let $m = qd + r$, with $0 \leq r \leq d-1$ and $q \in \mathbb{N}$. Then, $\lambda_m = \left(\prod_{i=0}^{d-1} w_i\right)^q = \rho^q$. Hence,

$$\|W^{(m)}z\| = \lambda_m \cdot \|W^{(r)}z\| \leq \rho^q \cdot \max_{0\leq r<d} \|W^{(r)}z\| = \rho^{\lfloor m/d \rfloor} \cdot C,$$

where $C := \max_{0\leq r\leq d-1} \|W^{(r)}z\|$ depends only on z and d. Since $\rho < 1$, it follows that $\|W^{(m)}z\| \to 0$ exponentially as $m \to \infty$. ■

Proof of (17). Fix a tolerance $\varepsilon > 0$, let $\rho = \prod_{i=0}^{d-1} w_i$ and the effective horizon

$$M_{\text{eff}}(x;\varepsilon) := \min\{M \geq 0 : \|J_k W^{(m)}x\| \leq \varepsilon,\ \forall m \geq M\}.$$

Let $C := \max_{0\leq r\leq d-1} \|J_k W^{(r)}x\|$, by Lemma 35, a sufficient condition for $\|J_k W^{(m)}x\| \leq \varepsilon$ is $C\,\rho^{\lfloor m/d \rfloor} \leq \varepsilon$, or equivalently $\rho^{\lfloor m/d \rfloor} \leq \frac{\varepsilon}{C}$. Since $0 < \rho < 1$, the function $x \mapsto \log_\rho x$ is decreasing; thus, $\lfloor m/d \rfloor \geq \frac{\log(\varepsilon/C)}{\log \rho}$. Therefore,

$$m \ \geq\ d\left\lceil \frac{\log(\varepsilon/C)}{\log \rho} \right\rceil. \tag{20}$$

We have proved that if m satisfies (20), then $\|J_k W^{(m)}x\| \leq \varepsilon$. Thus, we arrive at the explicit upper bound $M_{\text{eff}}(x;\varepsilon) \ <\ d\left\lceil \frac{\log(\varepsilon/C)}{\log \rho} \right\rceil$. This completes the proof of (17). ■

Table 7: Experiment 1: Phase–3^{real} MP/CCDS behavior under deviation–control ablation on the weak–alignment synthetic dataset (10% scaffold budget, $d = 10$, $M = 700$, operator family ILL–SVD).

γ_{dev}	$\varepsilon^{\text{fast}}_{\text{param}}$	$\varepsilon^{\text{full CCDS}}_{\text{param}}$	Mean steps (fast)	Realizable (%)	Dominant stop
0.0	1.41×10^{-7}	8.54×10^{-2}	20.5	36.71	max_steps
0.5	1.38×10^{-7}	8.54×10^{-2}	25.8	36.43	max_steps
0.9	8.62×10^{-3}	8.54×10^{-2}	50.0	36.57	max_steps

Table 8: Experiment 2: Interpretability metrics for the fast MP/CCDS solver.

Operator	Overall Sparsity (%)	Nonzero Coeffs (total)	Label-Conditional Top Atoms (fast solver)	Key Interpretability Note
I	97.01	26,868	323 (29–32% in most classes), 329 (20–28%)	Shared vocabulary across classes
ORTHO	97.02	26,795	323 (25–32%), 329 (23–27%), 443 (20–26%)	atom alternation in hard cases
ILL_SVD	97.03	26,753	323 (20–27%), 329 (22–27%), 317 (19–22%)	class-biased dominant atoms
LOWRANK	97.05	26,566	323 (19–31%), 443 (18–24%), 329 (19–25%)	interpretable low-dim projs
SPIKED	97.00	26,986	323 (27–33%), 329 (22–26%), 443 (18–24%)	Spikes amplify dominant directions
SPARSE	97.03	26,745	323 (27–30%), 329 (23–28%), 317 (20–23%)	sparsity aligns with operator
TOEPLITZ	97.02	26,795	329 (25–30%), 323 (24–27%), 325 (18–24%)	Structured reuse patterns
ADV_SCAFF	97.02	26,838	323 (26–32%), 329 (20–28%), 325 (21–26%)	Robust features despite misalignment

Table 9: Experiment 3: Scaffold-quality diagnostic on the synthetic realizability benchmark. Better scaffold quality, measured through lower projection residual and higher explained label-space energy, correlates with lower parametrization error and higher realizability success. Values are mean ± standard deviation over runs.

Scaffold type	$\varepsilon_{\text{proj}} \downarrow$	$E_{\text{expl}} \uparrow$	$\varepsilon_{\text{param}} \downarrow$
Selected orbit	0.29 ± 0.04	0.84 ± 0.03	0.018 ± 0.006
Random orbit	0.47 ± 0.06	0.62 ± 0.05	0.071 ± 0.021
Gaussian	0.61 ± 0.07	0.45 ± 0.06	0.116 ± 0.028

Table 10: Discovery-budget ablation. Gains are strongly front-loaded: increasing the discovery fraction from 1% to 3% and 5% substantially improves both predictive accuracy and explained label-space energy, while the gains at 15% and 25% are marginal, indicating early recovery of the relevant orbit geometry and saturation of scaffold quality.

Dataset	Metric	1%	3%	5%	15%	25%
Gas Sensor Drift	Accuracy (%) $\uparrow$	65.9	69.4	70.3	70.7	70.8
Gas Sensor Drift	$E_{\text{expl}} \uparrow$	0.68	0.77	0.80	0.82	0.82
MNIST	Accuracy (%) $\uparrow$	93.4	95.7	96.1	96.3	96.3
MNIST	$E_{\text{expl}} \uparrow$	0.74	0.84	0.87	0.88	0.88